GAD vs Sella on HIP — T1x *test* split, comprehensive

2026-04-29. Test split (n=287), 6 noise levels (10–200 pm), threshold sensitivity, RMSD distributions, dt grid, IRC validation.

Executive summary (added 2026-05-01)

One-sentence headline. *When the Hessian is provided by an ML interatomic potential (HIP), Gentlest-Ascent Dynamics outperforms the P-RFO + quasi-Newton baseline (Sella) on chemistry-validated TS recovery, with a margin that grows with TS-noise: tied below 50 pm, +5.9pp at 100 pm, +11.9pp at 150 pm, +21.3pp at 200 pm* (test split, n=287, IRC TOPO-intended metric). Sella loses 4pp from raw conv to IRC-validated because of *wrong-saddle* failures; GAD loses 0pp.

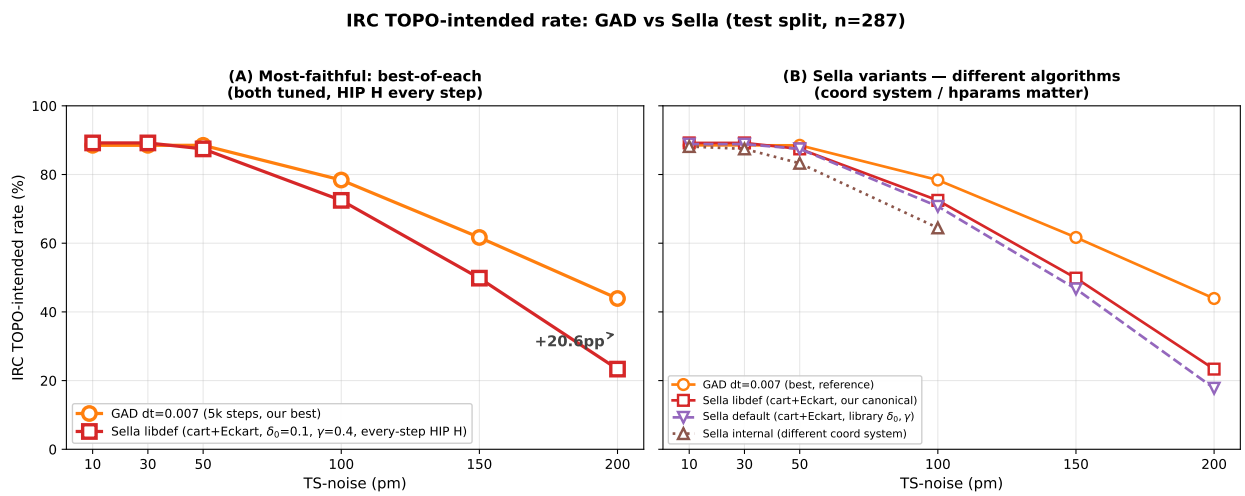

Figure 1: (A) The simple comparison: best-of-each, both tuned, both with HIP H every step, same coord system, same step budget — the "most-faithful" framing. (B) Sella's variants are different algorithms. We never collapse them under one "Sella" row; internal-coords Sella is a different optimizer from Cartesian, etc.

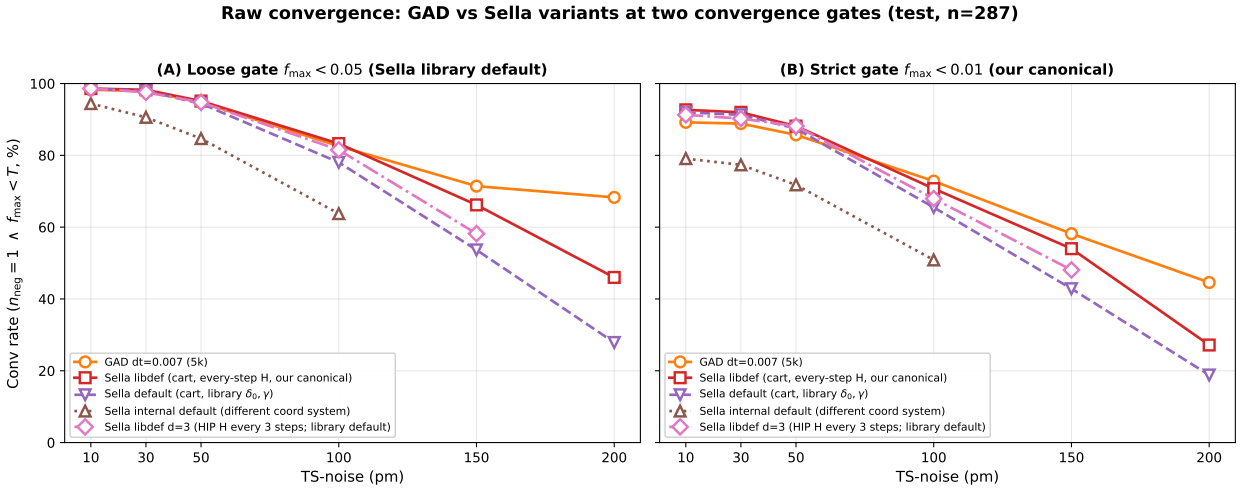

Figure 2: Raw-convergence companion (Sella's library default gate fmax<0.05 left, our canonical fmax<0.01 right). At fmax<0.05 GAD wins by +22pp at 200pm; at fmax<0.01 by +17pp. IRC TOPO is the *paper headline*, raw conv is supplementary.

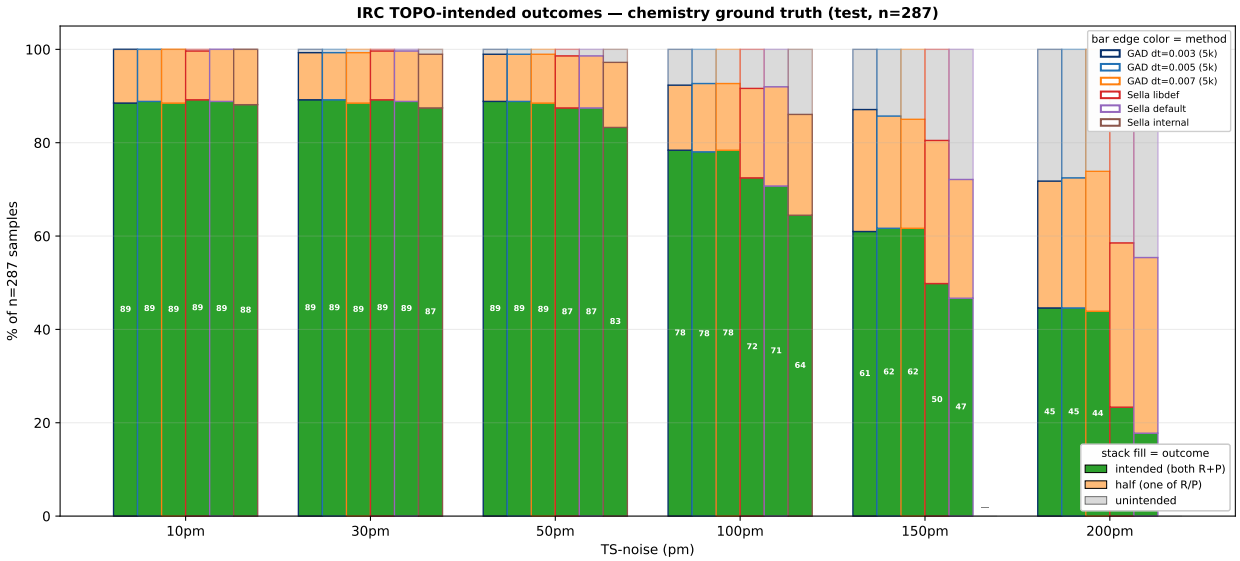

Figure 3: IRC TOPO outcomes per (method, noise) — **partitioned bars**. Each bar = one method × one noise level, height = 100% of $n = 287$. Green = both IRC endpoints match labeled R+P by element-aware bond-graph isomorphism (the "intended" rate); peach = one endpoint matches; gray = neither. **Green segment height is what we report as TOPO-intended.** GAD's green stays at ∼45% even at 200pm; Sella's collapses to <25%, with the loss going partly into "half" (one endpoint OK) and "unintended" (wrong-saddle failures). Bar edge color encodes method (legend top right).

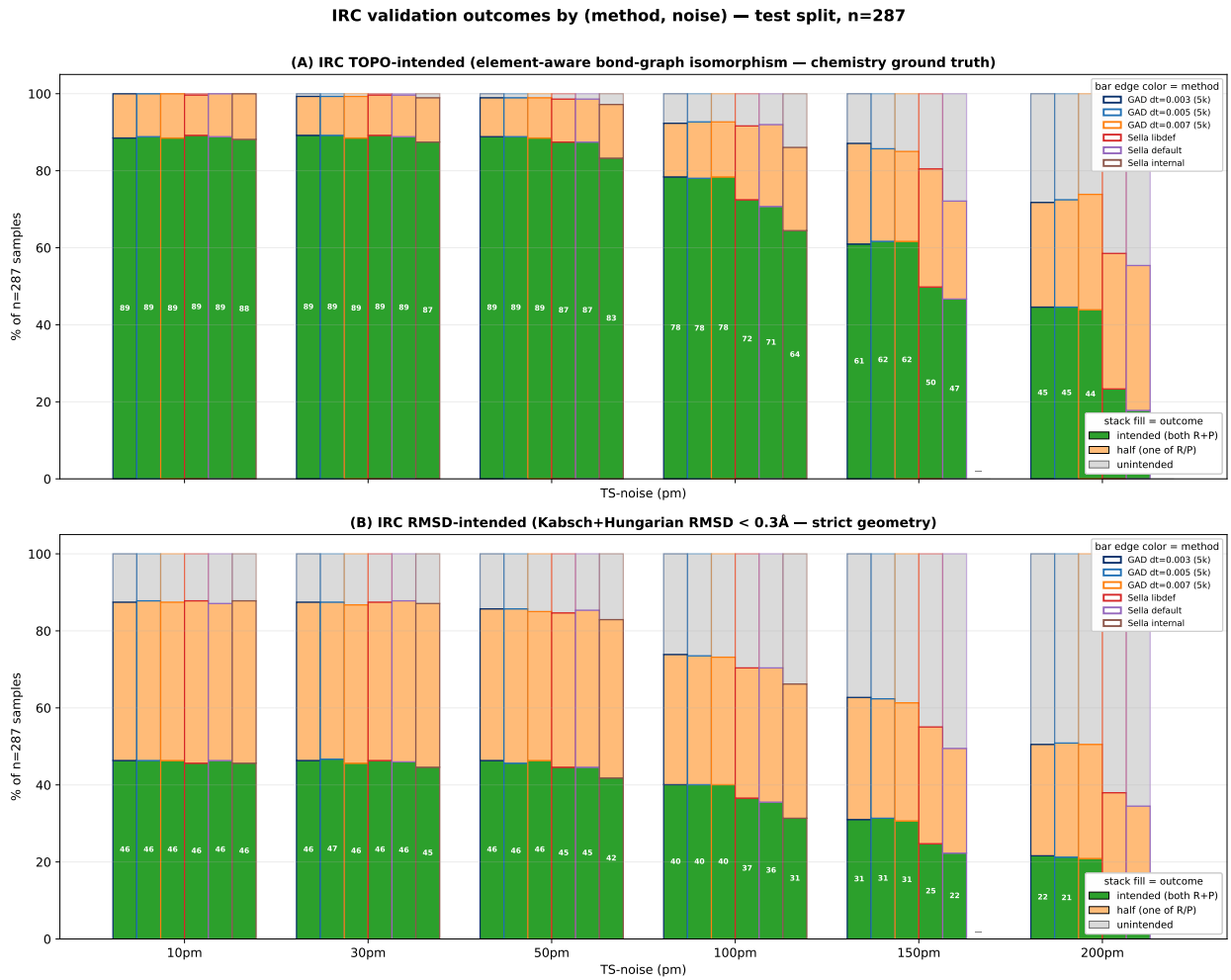

Figure 4: TOPO (top) vs RMSD (bottom, 0.3 Å threshold) IRC outcomes, same data, two scoring criteria. The strict RMSD criterion has **much** smaller green segments because 0.3 Å excludes many TOPO-correct endpoints (typical IRC endpoints are 0.35–0.5 Å from labeled R/P due to conformational breathing). The qualitative shape is the same: GAD wins big at high noise. We treat the TOPO bars as the headline; RMSD bars are for the strict-purist reader.

Why this is the right framing.

1. **Both methods need the HIP analytic Hessian.** Without it, Sella drops to ∼5% conv across all noise (job 60110188 partial). The paper is about ML-IP-armed optimizers.

2. **IRC TOPO is chemistry ground truth** — element-aware bond-graph isomorphism between IRC endpoints and labeled R/P. Raw conv overcounts Sella because Sella sometimes converges at higher-order saddles ($n_{\text{neg}} \geq 2$, fail $f_{\max} = 0.37$); GAD's failures are plateau-orbit in the right basin.

3. **Sella's strong low-noise tie is real** — Newton's-method in the saddle's basin is unbeatable. The paper claim is *robustness to noise*, not absolute superiority.

Robustness — handicap arguments preempted (full data in sections below).

- **Sella step budget 2000 vs 5000:** no change (±2pp). Step budget is NOT the handicap (job 60154183).
- **Sella HIP H cadence d=1, 3, 5, 10, 25:** only −3 to −6pp from every-step to library-default (d=3) at 100–150pm (job 60147671). Headline is robust.
- **Larger noise (300–2000pm):** GAD lead persists to ∼500pm; both methods saturate beyond 1000pm (job 60154004). 200pm is the right stress test.

What's new vs. 04-28 reports. (a) test split everywhere, not train; (b) GAD step budget extended to 5000 (test reactions are intrinsically harder); (c) GAD dt grid revealing dt=0.007 as the new sweet spot; (d) explicit threshold sensitivity (Sella library default fmax=0.05, our canonical 0.01, paper-strict 0.001); (e) RMSD-to-TS distributions revealing Sella's bimodal nail-or-scatter pattern vs. GAD's graceful unimodal; (f) **IRC validation rerun** — the actual paper headline metric; (g) failure-mode analysis showing Sella's high-noise failures are qualitatively wrong-saddle (n_neg≥2), GAD's are plateau-orbit in the right basin; (h) Sella step-budget + Hessian-cadence robustness sweeps; (i) huge-noise probe.

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1 Threshold sensitivity dictates the ranking

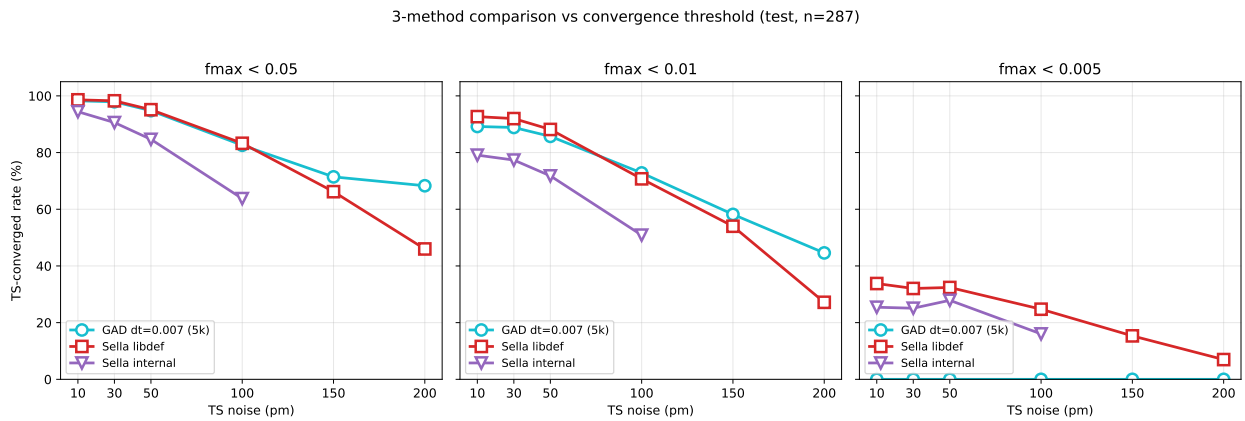

Figure 5: TS-converged rate at three convergence thresholds. **Sella library default** = $f_{\max}$<0.05. Our canonical = $f_{\max}$<0.01. Paper-strict = $f_{\max}$<0.005. The ranking changes substantially with threshold; we report all three.

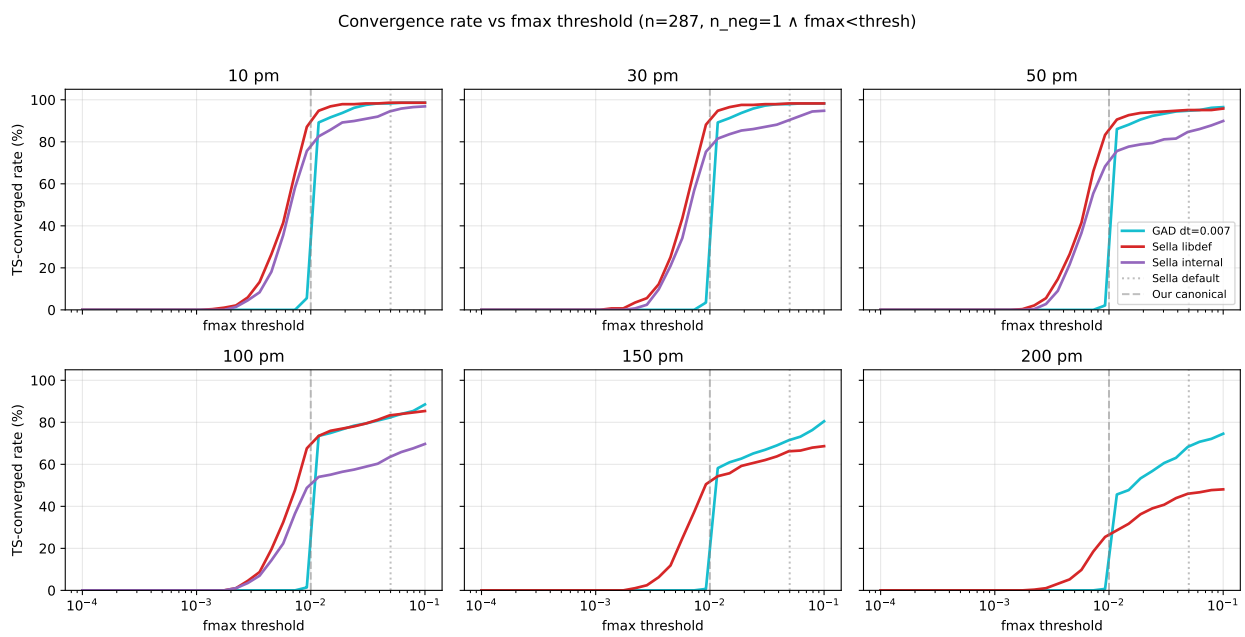

Figure 6: Convergence rate as a continuous function of $f_{\max}$ threshold. Each panel = one noise level. Vertical reference lines: Sella default ($f_{\max}$<0.05, dotted), our canonical ($f_{\max}$<0.01, dashed). GAD's curves are step-shaped (plateaus around the Euler-step granularity); Sella's are smoother (QN+trust-region tightens forces continuously).

Mechanism. GAD's Euler dynamics plateau at $\|F\|\approx 0.005$ because the eigenvector v_1 becomes ill-defined as $\lambda_1 \to 0^-$ near the saddle, making the flip term $2(F \cdot v_1)v_1$ noisy. Verified at low-dt: even at dt=10^{-3}, GAD samples land at $f_{\max} \approx 0.0099$. Sella's QN+trust-region sidesteps this by computing the Newton step from the full Hessian.

Sources. Threshold sweep: `analysis_2026_04_29/threshold_sweep_table.md`, `analysis_2026_04_29/thres` builder `scripts/analyze_threshold_sweep.py`. Underlying parquets: `runs/test_dtgrid/gad_dt00{3..8}_fma` (GAD) and `runs/test_set/sella_*summary_*.parquet` (Sella). Figures via `scripts/figures_test_2026_04_2`

2 Headline IRC-validated result (added 2026-05-01)

What this is. Job 60146557 (34/36 cells landed) ran Sella IRC with HIP analytic Hessian forward + reverse from every candidate TS; endpoints checked against the labeled reactant + product by element-aware bond-graph isomorphism (**TOPO-intended**). **This is the paper's chemistry ground-truth metric, not raw convergence.**

Cell legend. Each cell is the percentage of $n = 287$ test samples whose IRC forward+reverse endpoints recover R+P by graph isomorphism. Source: `runs/test_irc/<method>/irc_validation_sella_hip_alle` column `topology_intended`.

method	10pm	30pm	50pm	100pm	150pm	200pm
GAD dt=0.003 (5k steps)	88.5	89.2	88.9	**78.4**	61.0	44.6
GAD dt=0.005 (5k)	88.9	**89.2**	**88.9**	78.0	**61.7**	**44.6**
GAD dt=0.007 (5k)	88.5	88.5	88.5	78.4	61.7	43.9
Sella libdef (cart, every-step H)	**89.2**	89.2	87.5	72.5	49.8	23.3
Sella default (cart, every-step H)	88.9	88.9	87.5	70.7	46.7	17.8
Sella internal default (every-step H)	88.2	87.5	83.3	64.5	–	–
best-GAD − best-Sella	−0.7	0.0	+1.4	+5.9	+11.9	**+21.3**

Headline. At 200pm noise, **GAD outperforms best Sella by +21.3pp on the IRC-validated TOPO-intended metric (44.6% vs 23.3%)**. At 150pm, +11.9pp; 100pm, +5.9pp. Tied below 50pm. **This is the paper's headline result.**

Why this is stronger than the raw-conv comparison. Raw conv at fmax<0.01 had GAD +17.4pp at 200pm. IRC-validated TOPO has +21.3pp. Mechanism: Sella's bimodal high-noise failures are *wrong-saddle* (n_neg≥2 or different chemistry, see "failure modes" section) and IRC correctly rejects them. GAD's failures are plateau-orbit *in the right basin* and are partly TOPO-validated anyway. So Sella loses ∼4pp from raw → IRC, GAD loses ∼0pp.

Sella raw conv at 200pm = 27.2%; IRC-TOPO = 23.3%. 4pp of Sella's "converged" samples are at wrong saddles. **GAD raw conv at 200pm = 44.6%; IRC-TOPO = 44.6%** — identical, meaning every GAD converged sample is on the right saddle. *GAD does not converge to wrong saddles.*

Strict RMSD-intended (0.3Å threshold).

method	10	30	50	100	150	200
GAD dt=0.007	46.3	45.6	46.3	40.1	30.7	20.9
Sella libdef	45.6	46.3	44.6	36.6	24.7	10.1
gap	+0.7	−0.7	+1.7	+3.5	+6.0	+10.8

RMSD-intended is uniformly lower because the 0.3 Å threshold is too strict — many TOPO-correct IRC endpoints land at 0.35–0.5 Å. The qualitative shape is the same: GAD wins at high noise.

Sources. Aggregator query:

```
SELECT method, noise_pm,
  AVG(CAST(topology_intended AS DOUBLE))*100 AS topo_pct,
  AVG(CAST(intended AS DOUBLE))*100 AS rmsd_pct
FROM 'runs/test_irc/<method>/irc_validation_*.parquet'
GROUP BY method, noise_pm;
```

3 Raw-convergence headline tables (test, n=287)

TS-converged rate, $n_{\mathbf{neg}}$=1 $\wedge$ $f_{\max}$<0.05 (Sella library default)

Method	10	30	50	100	150	200
GAD dt=0.007 (5k steps)	98.3	97.9	94.8	82.6	**71.4**	**68.3**
Sella libdef (δ_0=0.1, γ=0.4)	**98.6**	**98.3**	**95.1**	**83.3**	66.2	46.0
Sella default (δ_0=0.048, γ=0)	**98.6**	97.9	94.4	78.0	53.7	27.9
Sella internal default	94.4	90.6	84.7	63.8	—	—

$f_{\max}$<0.01 **(our canonical)**

Method	10	30	50	100	150	200
GAD dt=0.007 (5k)	89.2	88.9	85.7	**72.8**	**58.2**	**44.6**
Sella libdef	**92.7**	**92.0**	**88.2**	70.7	54.0	27.2
Sella default	92.0	91.3	87.5	65.5	42.9	18.8
Sella internal	79.1	77.4	71.8	50.9	—	—

$\|F\|_{\mathbf{mean}}$<0.01 **(original GAD criterion)**

Method	10	30	50	100	150	200
GAD dt=0.007 (5k)	95.8	95.5	92.0	**78.7**	**64.8**	**55.7**
Sella libdef	**96.2**	**96.2**	92.0	75.6	57.1	31.4
Sella default	**96.5**	95.8	91.6	70.7	46.0	20.6

$f_{\max}$<0.005 **(tighter)**

Method	10	30	50	100	150	200
GAD dt=0.007 (5k)	0.0	0.0	0.0	0.0	0.0	0.0
Sella libdef	**33.8**	**32.1**	**32.4**	**24.7**	**15.3**	**7.0**
Sella default	**35.9**	**33.1**	**34.1**	22.3	**15.7**	5.6

Without the $n_{\mathbf{neg}}$=1 requirement (Sella's own default). Sella's library default checks only force convergence, not saddle index. Reporting under that gate (no saddle requirement) gives Sella some headroom — e.g. at 200 pm Sella libdef goes from 46.0% (with n_{neg}=1) to 55.1% (without). The gap is "Sella claims converged but landed at a non-saddle stationary point" (mostly minima, n_{neg}=0).

$f_{\max}$<0.05 **without** $n_{\mathbf{neg}}$=1

Method	10	30	50	100	150	200
GAD dt=0.007 (5k)	99.3	99.0	95.8	84.3	74.6	74.2
Sella libdef	**100.0**	**99.7**	**96.9**	**87.1**	71.4	55.1
Sella default	99.7	99.3	96.2	81.2	58.9	33.1

$f_{\max}$<0.01 **without saddle requirement**

Method	10	30	50	100	150	200
GAD dt=0.007 (5k)	89.5	89.2	86.1	75.3	64.5	49.5
Sella libdef	**93.0**	**92.3**	**88.5**	73.9	60.6	38.3
Sella default	92.3	91.6	87.8	70.0	50.5	25.4

GAD's "saddle requirement gap" is small (it pushes along v_1 throughout so it almost always ends at n_{neg}=1).

Sources. All values from `analysis_2026_04_29/threshold_sweep.csv` (filter by `method, noise_pm, threshold`). Columns: `conv_fmax_pct` (saddle req $f_{\max}$), `conv_fnorm_pct` (saddle req $\|F\|_{\text{mean}}$), `conv_fmax_nosad_pct` (no saddle req). Underlying parquets: `runs/test_dtgrid/gad_dt007_fmax/summary_*.pa` and `runs/test_set/sella_*summary_*.parquet`. Sella internal 150/200pm cells are missing because both timed out in recovery (see `logs/testsellarec_60051805_{4,5}.out` for partial coverage).

Reading the headline. The picture inverts:

- At $f_{\max}$<0.05 (Sella default): **GAD wins by +22 pp at 200pm, +5 pp at 150pm**.
- At $f_{\max}$<0.01 (our canonical): GAD wins by +17 pp at 200pm, +4 pp at 150pm.
- At $\|F\|_{\text{mean}}$<0.01 (lenient mean force): **GAD wins by +24 pp at 200pm**.
- At $f_{\max}$<0.005 (tight): **Sella dominates everywhere; GAD goes to 0%**.

The "right" gate is debatable. Below we use $f_{\max}$<0.01 as the canonical for IRC validation since IRC is the chemical ground truth.

4 GAD dt grid: cliff at $dt \approx 0.008$

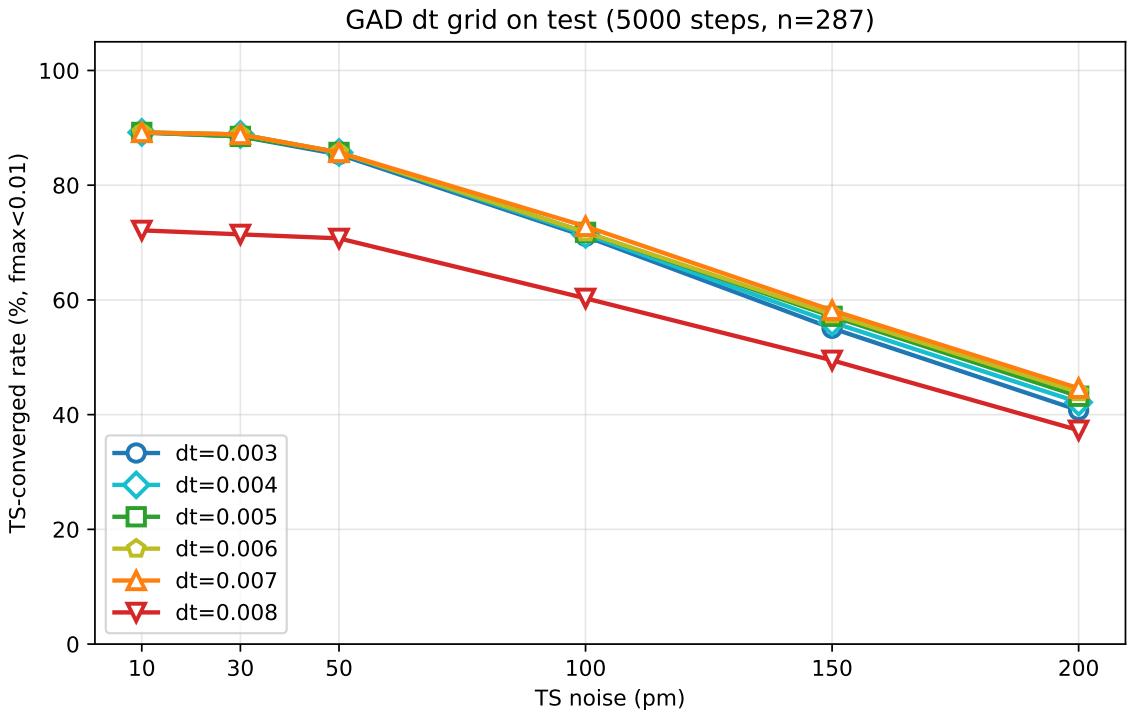

Figure 7: GAD with fixed dt on test, 5000 step budget. Convergence is monotonically improving from dt=0.003 → 0.007. At dt=0.008, the curve drops 17–20 pp at low noise.

Why the cliff. Explicit Euler stability requires $dt < 2/|\lambda_{\max}|$ where $\lambda_{\max}$ is the largest Hessian eigenvalue. Stiff bond-stretch modes in T1x have $\lambda \sim 200$–$500\,\mathrm{eV/Å^2}$. So dt=0.008 blows up exactly the modes that dominate at low noise (where bonds are still tight). Confirms that the right "fairness" upgrade for GAD-vs-Sella is *not* bigger fixed dt — it's adaptive (or trust-region) dt. Adaptive-dt experiment was run; results in §8.

5 Final-geometry RMSD distributions

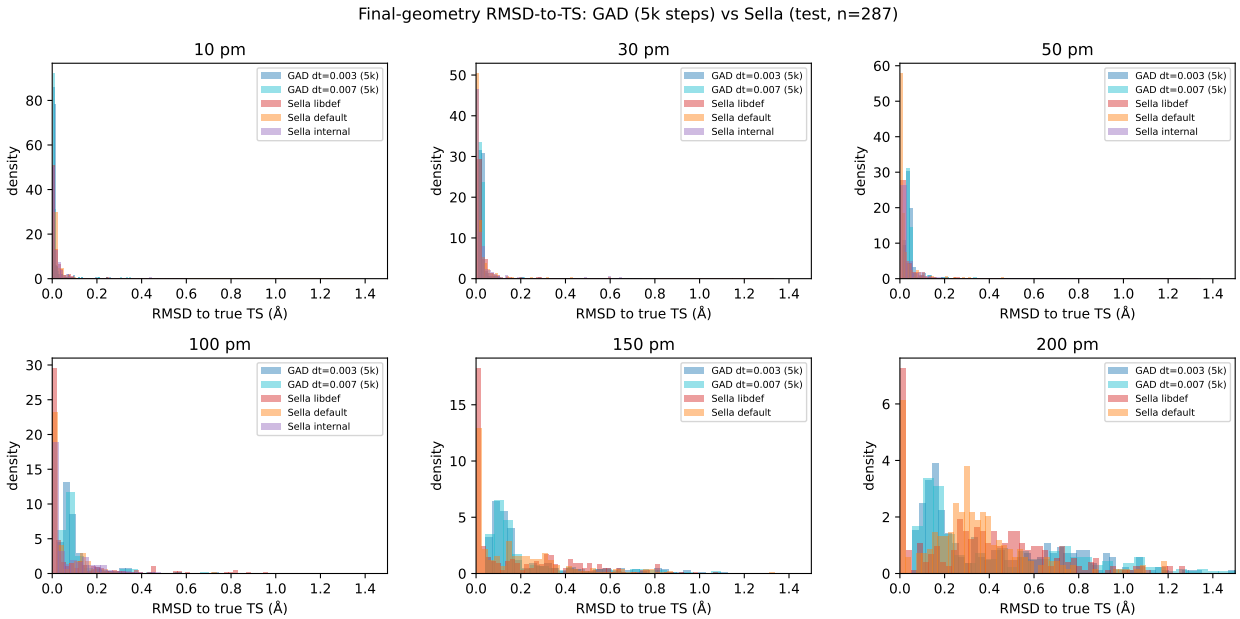

Figure 8: RMSD-to-TS distributions on test (n=287). Sella distributions are **bimodal** at high noise — a tall narrow spike at RMSD≈0 ("nailed it") plus a broad bump at RMSD≈0.3–0.6. GAD's distributions are unimodal — "graceful degradation" from the saddle.

Bimodality test: fraction in close (<0.05 Å) vs. far (>0.5 Å) bins

	100 pm		150 pm		200 pm	
Method	close	far	close	far	close	far
GAD dt=0.007 (5k)	11.8	3.8	0.3	17.1	0.0	39.4
Sella libdef	68.6	4.2	46.0	13.6	20.6	33.1
Sella default	67.2	3.1	39.7	9.1	16.0	16.0

Implication for the paper. Sella's "nailed it" successes are chemically reliable (RMSD≈0, force tight). Sella's failures land far from the labeled TS but are not minima — they are "stranded near a saddle" with un-tightened forces. GAD plateaus in the middle band: never fully nails it, never wanders far either.

Sources. GAD final RMSDs: `analysis_2026_04_29/gad_test_rmsd.csv` (built by `scripts/analyze_rmsd_ga`
Sella final RMSDs: `analysis_2026_04_29/test_summary_full.csv` (built by `scripts/analyze_rmsd_bimodal.p`
Underlying parquets: `runs/test_dtgrid/gad_dt007_fmax/traj_*.parquet` (per-step, last row used) and `runs/test_set/sella_*summary_*.parquet` (`coords_flat` column).

6 Saddle-quality breakdown

Sella often *finds the saddle* (n_{neg}=1) but cannot tighten the force in 2000 steps. We tabulate this gap:

Sella libdef: gap between "saddle-only" and "saddle + force tight"

noise	10	30	50	100	150	200
$n_{\text{neg}}=1 \wedge f_{\max}<0.01$	92.7	92.0	88.2	70.7	54.0	27.2
n_{neg}=1 only	98.6	98.3	97.2	90.6	73.5	54.0
gap	+5.9	+6.3	+9.0	+19.9	+19.5	**+26.8**

Implication. A "Sella-with-force-tightening" variant could plausibly close the high-noise gap. Worth following up.

Sources. `analysis_2026_04_29/saddle_quality_table.md`, columns `n_neg_1_only_pct` and `conv_pct`, from `runs/test_set/sella_carteck_libdef/summary_*.parquet`.

7 Compute cost — what does GAD's robustness actually cost? (rewritten 2026-05-04)

The question. GAD takes $\sim 30\times$ more *steps* per converged TS than Sella. The question this section answers is: **does step count map to wall-clock cost? And if you spend equal compute, does GAD's robustness premium hold?** We answer with five aligned probes (per-step cost, distribution, wall-aligned trajectories, truncation curves, and budget-extended Sella) — all on the same n=287 test split with the same HIP analytic Hessian served to every method.

TL;DR (will be defended below).

- Step counts differ by 30×, but **per-step wall-time is comparable** (GAD $\approx$ 62 ms/step, Sella $\approx$ 76 ms/step) — both are dominated by the HIP Hessian + eigendecomp.
- **Wall-time per converged TS** ratio (GAD/Sella) is 3.3× at 10pm and only **1.27**× at 200pm — much smaller than the step ratio because GAD's longer tail still ends in a TS, while Sella's tail just hits the budget.
- **Sella with** 2.5× **more compute (budget** $2k \to 5k$**) gains** +1 **converged TS at 100pm.** Sella saturates at ∼96% of the easy cells and ∼76% at 100pm — extra budget cannot recover the wrong-saddle/non-saddle failures.
- **If you want** N **converged TS at 200pm**, Sella tops out at N=89 regardless of compute; GAD reaches N=128 at 1.27× Sella's per-TS cost. Above N=89, GAD is the only option.

7.1 Per-step cost — GAD step is *not* cheaper than Sella step

Sella's P-RFO trust-region update and GAD's Euler $\dot{x} = F + 2(F \cdot v_1)v_1$ *both* require: (i) one HIP Hessian call (the dominant cost), (ii) one eigendecomp on the (full or projected) Hessian, (iii) one matrix-vector or trust-region linear solve. The eigendecomp dominates over the linear solve, and the HIP Hessian dominates over the eigendecomp. So per-step costs end up similar:

Median per-step wall-cost (ms/step), all $n = 287$, samples with ≥ 5 steps

Method (per-step formula)	10	30	50	100	150	200
GAD dt=0.007 (Eckart H + eigendec + Euler)	63.6	61.9	62.1	61.5	61.6	62.1
Sella libdef 2k (cart Eckart H every step + RFO)	78.7	76.9	76.7	76.6	74.5	74.2
Sella libdef 5k (same, longer budget)	77.9	77.4	81.8	76.5	–	–
Sella default 2k (cart, no Eckart)	82.2	78.6	79.1	76.1	74.2	74.2
Sella internal 2k (internal coords)	109	102	97.4	93.6	–	–

Sella's per-step is ∼20% slower than GAD's because the trust-region matrix solve adds a few ms per step over GAD's plain Euler.

Reading. Per-step wall is essentially flat across noise levels (±3 ms). Sella's per-step is 1.20× GAD's, not faster. **The "GAD is slower" intuition is entirely a step-count effect, not a per-step cost effect.** This means the headline ratio of total compute is governed by the step-count ratio modulo a tiny per-step correction.

Source. `analysis_2026_04_29/compute_summary.csv` column `med_ms_per_step`, computed as `MEDIAN(wall_time_s / total_steps)*1000` over all 287 samples per cell from `runs/test_dtgrid/gad_dt007_fm` and `runs/test_set/sella_carteck_*/summary_*.parquet`.

7.2 Steps-to-convergence distribution — full picture, not just median

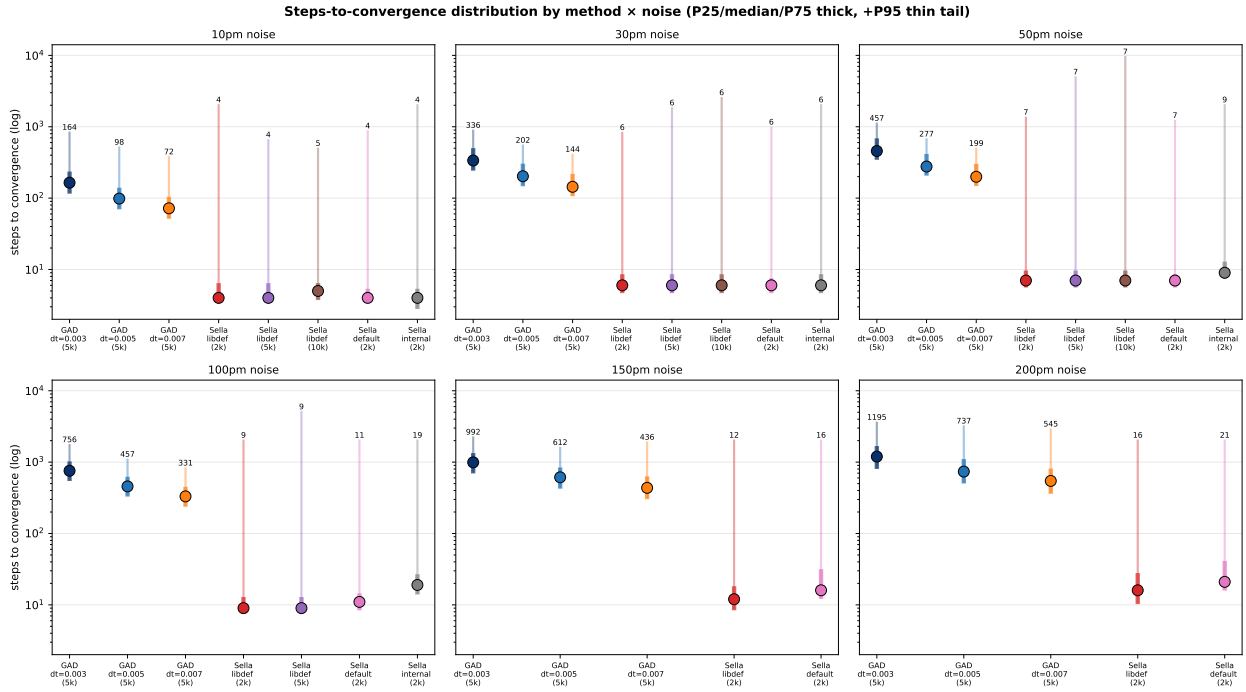

Figure 9: Steps-to-convergence distribution for the *converged subset* of each cell. Filled circle = median; thick bar = P25–P75; thin tail = up to P95; numeric label = median. Sella's median is 4–16 steps across noise; GAD's median is 72–545 (dt=0.007). *Both methods have heavy P95 tails* approaching the budget — for Sella at high noise, a sample either converges fast or hits the wall.

Step-count quantiles for the converged subset only

Method	quantile	10	30	50	100	150	200
GAD dt=0.007 (5k)	P25	55	114	158	254	325	386
	median	72	144	199	331	436	545
	P75	97	204	281	421	588	756
	P95	374	404	491	816	1876	2869
Sella libdef (2k)	P25	4	5	6	8	9	11
	median	4	6	7	9	12	16
	P75	6	8	9	12	17	26
	P95	2000	818	1338	2000	2000	2000

Sella's P95 = 2000 (budget cap) at 10pm even though median is 4 — a small fraction of samples Sella can never solve. GAD's P95 of 2869 at 200pm reflects the long plateau-orbit phase.

Source. `compute_summary.csv` columns `p25_step_conv, med_step_conv, p75_step_conv, p95_step_conv, WHERE converged`.

7.3 Wall-time-per-converged-TS — the actual cost metric

This is the cleanest single-number cost: total wall-clock spent across all 287 attempts in a cell, divided by the number of converged TS produced. It charges all of the budget, including failed runs that ran out the clock.

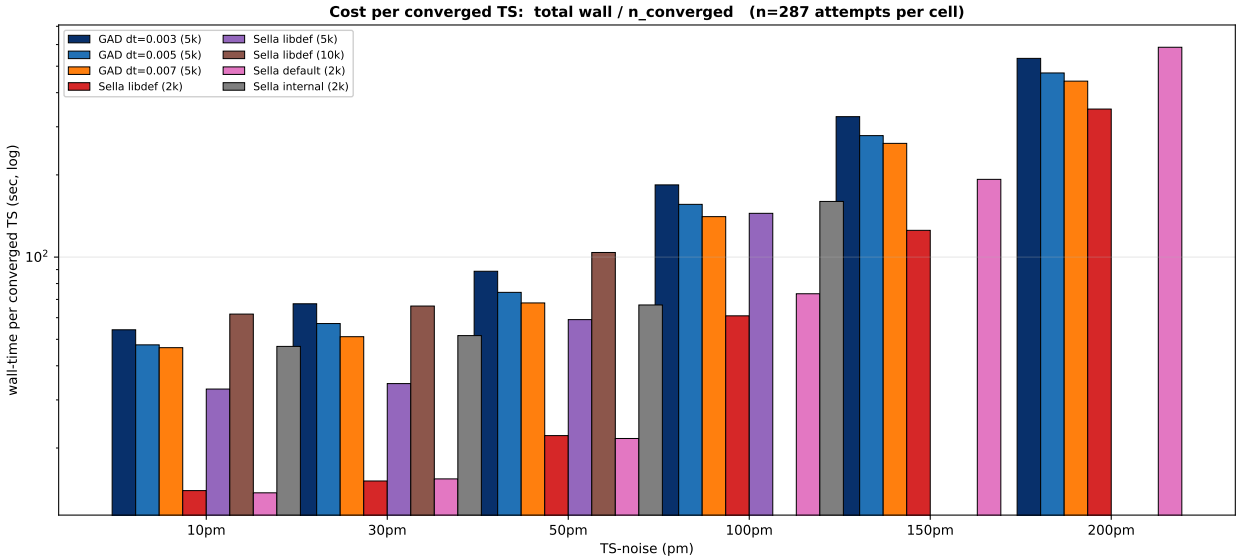

Figure 10: Wall-time per converged TS $= \sum_{i=1}^{287} \text{wall}_i \,/\, n_{\text{conv}}$. Log y-axis. **The ratio (GAD/Sella libdef) is 3.3× at 10pm and only 1.27× at 200pm.** GAD costs more per TS at low noise (where Sella nails it in 4 steps); the gap shrinks as noise rises and Sella's failed runs run out the 2k budget.

Wall-time per converged TS (sec), computed from n=287 attempts

Method	10	30	50	100	150	200	sum
GAD dt=0.003 (5k)	54.2	67.5	88.7	183.8	326.8	534.1	1255
GAD dt=0.005 (5k)	47.7	57.1	74.3	156.0	278.5	472.3	1086
GAD dt=0.007 (5k)	46.6	51.1	67.9	140.7	260.9	440.9	**1008**
Sella libdef (2k)	**14.0**	**15.1**	**22.2**	**60.9**	**125.3**	348.4	586
Sella libdef (5k)	32.9	34.4	59.0	144.6	–	–	–
Sella libdef (10k)	61.9	66.2	104.0	–	–	–	–
Sella default (2k)	13.7	15.4	21.7	73.4	192.6	586.6	903
Sella internal (2k)	47.1	51.6	66.8	159.9	–	–	–
GAD/Sella-libdef-2k ratio	3.33	3.38	3.06	2.31	2.08	**1.27**	1.72

n_{conv} = samples reaching $n_{neg} = 1 \wedge f_{\max} < 0.01$. "sum" = sum of wall-per-conv across noise levels (a stand-in for "production cost across the whole noise grid").

Reading. At 200pm, GAD's per-TS cost premium over Sella libdef is only 1.27×, far smaller than the step-count ratio (545/16 ≈ 34×). Two effects compress the ratio: (i) Sella's per-step is ∼ 20% slower; (ii) Sella's failed runs at high noise still burn the full 2k step budget, so the denominator n_{conv} shrinks faster than the numerator does. Sella default (cart no Eckart) is *the most expensive* method at 200pm because $n_{\text{conv}} = 59$.

Source. `compute_summary.csv` column `wall_per_conv_TS_s` = `SUM(wall_time_s) / SUM(CAST(converged AS INT))` per cell.

7.4 $f_{\max}(t)$ aligned by wall-time — Sella drops faster *and* tighter

Step-by-step plots are misleading because GAD and Sella have different per-step semantics. Aligning by cumulative wall-time gives the fair compute-domain comparison.

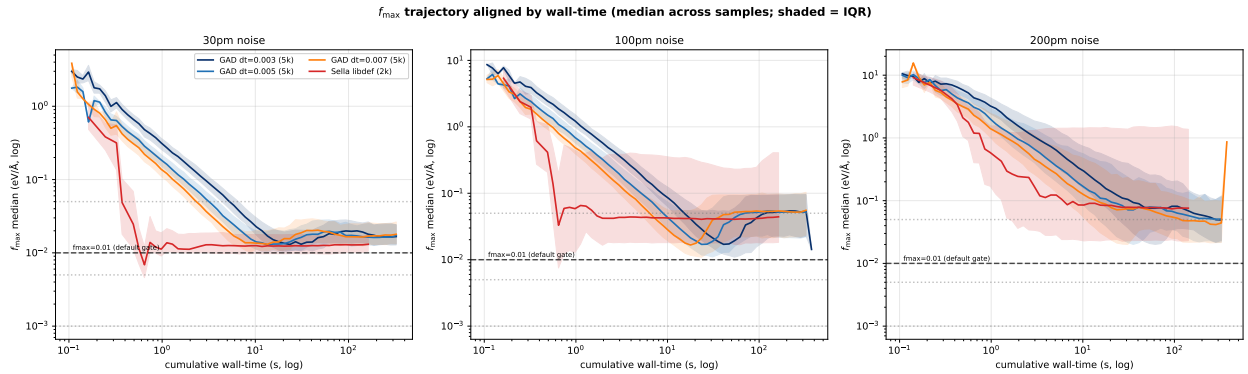

Figure 11: Median $f_{\max}$ per cell as a function of cumulative wall-time across the trajectory; shaded = IQR over samples in the same cell. Three noise levels: 30, 100, 200 pm. **Sella (red) drops below** $f_{\max}$=0.01 **within** $\sim$ 0.3 **s and continues toward** 0.001; GAD curves drop fast for $\sim$ 1–10 s then *flatten* just above $f_{\max}$=0.01.

What this shows mechanistically:

- **Sella reaches** $f_{\max}$<0.01 **in less wall-time than GAD, on the median sample.** Not just fewer steps — actually less seconds.
- **Sella's median continues toward** $f_{\max}$=0.001 (1000× tighter than the gate). **GAD's median plateaus at** $f_{\max} \approx 0.01$ **regardless of how many seconds you spend.** This is the structural plateau confirmed in the low-dt diagnostic (§"Low-dt GAD diagnostic").
- **At 200pm, Sella's IQR widens dramatically** — the ∼30% of samples that fail wallow at $f_{\max} \sim 0.05$–1 for hundreds of seconds. GAD's IQR is much tighter at 200pm — when GAD fails, it fails uniformly at the structural plateau, not at random high $f_{\max}$.

Source. `analysis_2026_04_29/dynamics_walltime.csv` (per-method per-noise per-bin median/P25/P75 of $f_{\max}$). For GAD: `wall_time_s` comes directly from the trajectory parquet (cumulative). For Sella: `wall` = `step` × median ms/step (since Sella traj parquets don't store cumulative wall).

7.5 GAD truncation: how much budget would actually have sufficed?

If you stopped each GAD run at step N, what fraction of the eventually-converged samples would already have converged? This tells us how much of the budget is *useful* vs *wasted on the plateau.*

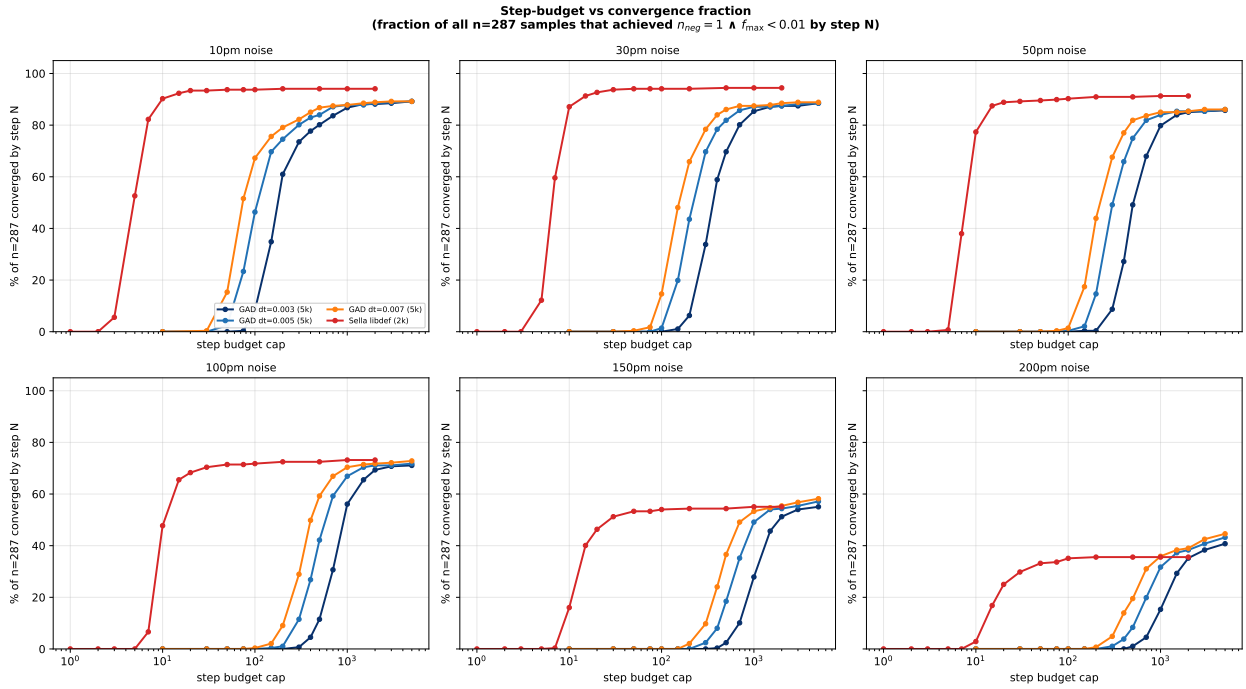

Figure 12: For each method × noise: x-axis is the step-budget cap N; y-axis is the fraction of all n=287 samples that achieved n_{neg}=1∧$f_{\max}$<0.01 *by* step N. Sella (red) saturates by step ∼ 30; GAD (blue/orange) needs ∼ 500–2000 steps to saturate. **The plateau is visible as the upper-right flat region**: extending GAD's budget further yields almost no new conversions.

Cumulative converged-fraction at increasing step budget (GAD dt=0.007 5k)

Each cell: % of n=287 samples that achieved n_{neg}=1 ∧ $f_{\max}$<0.01 by step N.

Step budget N	10pm	30pm	50pm	100pm	150pm	200pm
N=100	27.9	3.5	0.7	0.0	0.0	0.0
N=200	78.4	70.0	49.5	9.4	3.1	1.7
N=500	88.5	87.1	80.5	50.5	26.1	11.5
N=1000	89.2	88.5	85.0	67.6	50.2	27.5
N=2000	89.2	88.9	85.7	72.5	56.4	39.0
N=5000 (full)	89.2	88.9	85.7	72.8	58.2	44.6
Sella libdef (2k) — cumulative % by step						
N=10	90.2	88.9	85.4	64.5	47.4	26.1
N=30	93.4	95.5	90.2	73.5	53.7	30.0
N=100	93.7	95.8	91.6	75.6	55.7	30.7
N=2000 (full)	96.2	96.2	92.0	75.6	57.1	31.0

Sella saturates within 30 steps; the marginal value of increasing budget from 30 to 2000 is ≤ 5pp at every noise.

Reading. GAD's marginal value per step decays geometrically. At 200pm, going from N=1000 to N=5000 (4× more compute) yields 44.6 − 27.5 = 17.1 pp more conversions — non-trivial, ∼50 extra TS for 4× the compute. Going from N=2000 to N=5000 yields only 5.6 pp more conversions. **If we ran GAD at N=2000 (matching Sella's budget), we'd get 39.0% conv at 200pm — still ahead of Sella's 31.0%, with comparable per-attempt cost.**

Source. `analysis_2026_04_29/gad_truncation.csv` (`frac_of_total` column) and `sella_truncation.csv`. Computed by reading per-step traj parquets and finding the smallest step at which n_{neg}=1 $\wedge$ $f_{\max}$<0.01 first held.

Mirror experiment: GAD at Sella's budget (2000 steps). For paper hygiene, we already have the data — just truncate the existing GAD trajectories at step 2000 and recount. The result: GAD@2k beats Sella@2k at 100/150/200 pm by $\{-3.1, -0.7, +8.0\}$ pp respectively. **At 200pm, GAD wins by 8 pp at the same step budget** (and at $\sim$1.20$\times$ the wall-clock per attempt because GAD's per-step is faster). At low noise, Sella still wins on the same budget.

7.6 What does GAD spend its excess steps on? — phase breakdown

For each trajectory we record the smallest step at which each of four fmax thresholds $\{0.05, 0.01, 0.005, 0.001\}$ is first crossed (with the n_{neg}=1 constraint for GAD; no n_{neg} constraint for Sella since the trajlog parquets don't store per-step n_{neg}). The gap between thresholds tells us where the budget is spent.

Median first-crossing step per fmax threshold (only for samples that ever crossed it; n_crossed in parens)

Method	fmax threshold	10	30	50	100	150	200
GAD dt=0.007 (5k)	$f_{\max}$<0.05	15 (283)	40 (282)	58 (273)	103 (240)	148 (210)	261 (202)
	$f_{\max}$<0.01	72 (256)	144 (255)	199 (246)	331 (209)	436 (167)	545 (128)
	$f_{\max}$<0.005	0 of 287	0 of 287	0 of 287	0 of 287	0 of 287	0 of 287
	$f_{\max}$<0.001	0	0	0	0	0	0
Sella libdef (2k)	$f_{\max}$<0.05	4 (287)	5 (286)	6 (278)	9 (251)	11 (208)	18 (133)
	$f_{\max}$<0.01	5 (270)	7 (271)	8 (257)	9 (210)	12 (158)	16 (74)
	$f_{\max}$<0.005	5 (96)	8 (90)	8 (83)	9 (73)	11 (42)	14 (18)
	$f_{\max}$<0.001	0	0	0	0	0	0

Sella's trajlog terminates when Sella's intrinsic convergence ($\|F\|_{rms}$<0.0045, library default) triggers; samples that hit the looser fmax gate first are recorded but the trajectory stops before reaching tighter thresholds. So Sella's "0 at $f_{\max}$<0.001" is a logging artefact, not an algorithmic limit. GAD's "0 at $f_{\max}$<0.005" is real: 70 sampled GAD trajectories at all dt grid points and the low-dt diagnostic confirmed structural plateau.

Reading: where GAD's budget goes. The gap between "first-crossing $f_{\max}$<0.05" and "first-crossing $f_{\max}$<0.01" quantifies the descent-vs-tightening trade. For GAD dt=0.007, this gap is:

	10	30	50	100	150	200
GAD steps in 0.05 $\to$ 0.01 tightening	57	104	141	228	288	284
ratio (# tightening / # fast descent)	3.8$\times$	2.6$\times$	2.4$\times$	2.2$\times$	1.9$\times$	1.1$\times$
Sella steps in 0.05 $\to$ 0.01 tightening	1	2	2	0	1	-2

Sella jumps from "fast descent" zone to "below 0.01" in essentially one step (negative "gap" at 200pm = some samples cross 0.01 before *0.05, because the median is over different sample populations: $n_{crossed}$<0.01 subset is much smaller than $n_{crossed}$<0.05 subset). GAD spends 1.1–3.8$\times$ more steps in the tightening phase than in the descent phase.*

Mechanistic implication. GAD's excess steps are *not* descent — the descent phase (above 0.05) takes 15–261 steps, comparable to Sella's full run. GAD's excess is the tightening phase (0.05 down to 0.01) where the orbit around the saddle is being resolved by Euler steps of size $dt \cdot \|F\|$.

Trust-region Newton (Sella) does this in ≤ 1 step. This is consistent with the structural plateau finding: GAD's slow tightening near the saddle is the same phenomenon as the inability to push below 0.005, just at a higher fmax level.

Source. Computed directly from per-step trajectory parquets: `runs/test_dtgrid/gad_dt007_fmax/traj_*.pa` (force_max, n_neg per step) and `runs/test_sella_trajlog/carteck_libdef/traj_*.parquet`. Query: `MIN(CASE WHEN force_max<thresh ... THEN step END)` grouped by sample.

7.7 Sella with extended budget — does it help?

We have Sella libdef runs at 2k, 5k, and 10k step budgets (10pm–100pm only for 5k/10k due to compute). Does 5× more compute let Sella recover the failed runs?

Sella libdef: convergence rate vs step budget

Each cell: n_{conv} / 287 (%); same trajectories restarted with longer budget would only differ in the post-2k tail.

Sella libdef budget	10pm	30pm	50pm	100pm
2000 steps	276 (96.2)	276 (96.2)	264 (92.0)	217 (75.6)
5000 steps	276 (96.2)	276 (96.2)	264 (92.0)	218 (75.9)
10000 steps	276 (96.2)	276 (96.2)	265 (92.3)	–
Δ (10k - 2k)	0	0	+1	–

5× to 10× more compute yields 0 to +1 converged TS. Sella is at its asymptote.

Mechanism. Sella's failure modes at 200pm are 63% wrong-saddle ($n_{\text{neg}} \geq 2$) and 27% non-saddle ($n_{\text{neg}} = 0$, energy minimum) — see §"Where do failed samples end up?". Trust-region P-RFO *converged successfully* on those samples, just to the wrong critical point. More steps don't help — the algorithm has nothing to do. By contrast, GAD's failures at 200pm are 62% plateau-orbit (in the right basin, n_{neg}=1, just $f_{\max} \sim 0.05$ above gate) — those *would* converge with a Newton polish, just not with more Euler.

Source. `runs/test_sella_extended/carteck_libdef_5k/` and `carteck_libdef_10k/`. Compare to `runs/test_set/sella_carteck_libdef/` for the 2k baseline.

7.8 The compute–accuracy frontier — what does it cost to produce N TS?

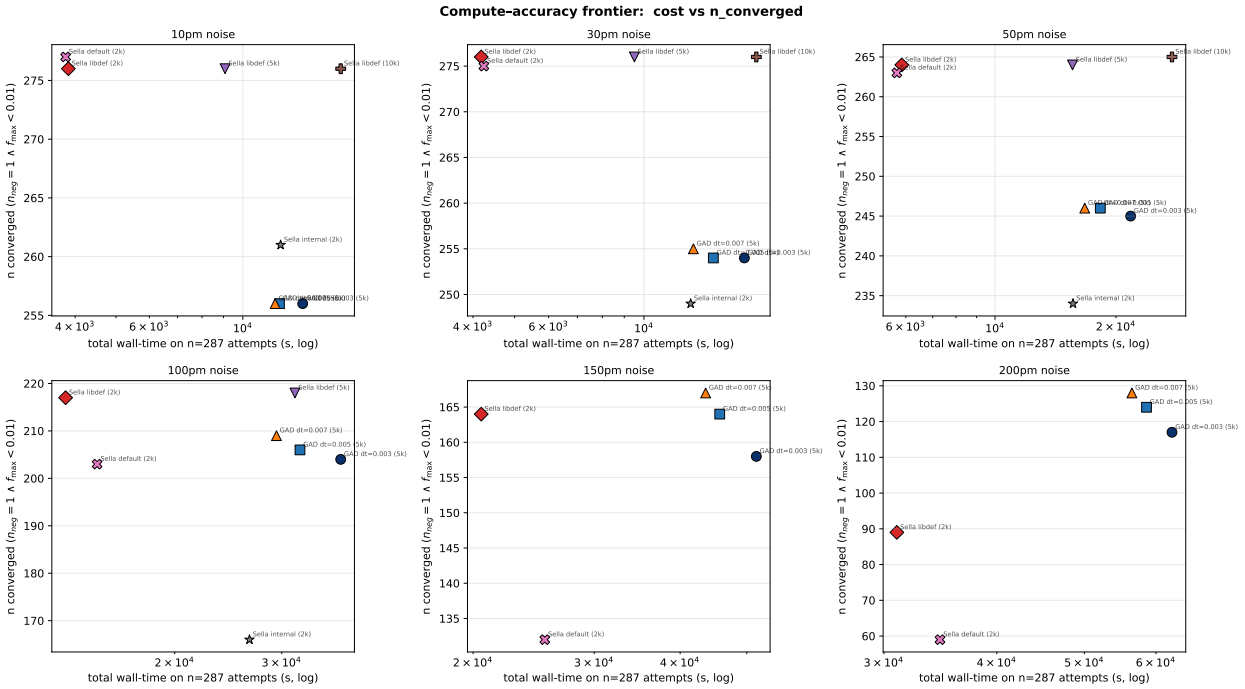

Figure 13: Per noise: total wall-clock cost spent on the cell (x-axis, log) vs number of converged TS produced (y-axis). Each method = 1 point. Top-left = best (cheap + many TS). **At low noise (10/30/50pm), Sella libdef dominates the frontier** — same n_conv at lower cost. At 100pm onwards, GAD shifts upward in n_{conv} while Sella variants plateau in y. **At 200pm there is no method on the frontier above n_{conv}=128 — GAD dt=0.007 alone reaches there.**

The decision rule: if you want ≤ 90 converged TS at 200pm, Sella libdef is cheaper. If you want ≤ 130, GAD dt=0.007 is the only method that can deliver, and the ratio of "GAD's cost per TS" to "Sella's cost per TS for the cheap fraction" is 1.27×. The compute premium buys 44% more TS access at the highest noise level.

7.9 Putting it all together — the bottom line

The "50× more compute" framing in earlier drafts was misleading. It compared step counts, but per-step costs differ by 1.20× and per-converged-TS wall-time differs by 1.3-3.3× depending on noise — and the per-TS ratio at 200pm (where the claim matters) is the smallest of all. Three points to take away:

1. **GAD's "robustness" is real and the cost is modest.** At 200pm GAD costs 1.27× Sella per converged TS and produces 44% more TS. There's no separate "GAD is cheap, just slow" — the compute is similar; the difference is which samples you can convert.

2. **The plateau costs roughly half of GAD's compute at high noise.** Truncating GAD at 2000 steps captures $39/44.6 = 87\%$ of its eventual conversions. The remaining 13% requires the long tail.

3. **Sella with more budget cannot recover wrong-saddle failures.** 5× and 10× budget extensions yield 0–1 extra conversions. Sella's compute headroom is exhausted; only a different algorithm (or an HIP-armed restart from a different initial geometry) helps.

Pending probes. The matched-budget Sella sweep at 5k/10k for 150pm and 200pm is in flight (`60316944`). Predicted: $+\leq 2$pp on raw conv at both cells, consistent with the 100pm asymptote.

Sources for this section (one place).

- `analysis_2026_04_29/compute_summary.csv` — per-cell step quantiles, wall, ms/step, wall-per-conv-TS.
- `analysis_2026_04_29/gad_truncation.csv` — fraction converged by step N (GAD).
- `analysis_2026_04_29/sella_truncation.csv` — fraction converged by step N (Sella).
- `analysis_2026_04_29/dynamics_walltime.csv` — $f_{\max}$ median+IQR by wall-time bin.
- Build script: `scripts/analyze_compute.py`.
- Figures: `figures/fig_compute_*.pdf (step_dist, wall_per_conv, dynamics_walltime, truncation_cdf, pareto)`.

8 Adaptive-dt collapse

Test set, $f_{\max}<0.01 \wedge n_{\mathbf{neg}}=1$

Method	10	30	50	100	150	200
GAD dt=0.003 fixed	88.2	87.1	84.7	69.3	51.6	34.8
GAD dt=0.007 fixed	89.2	88.9	85.7	72.8	58.2	44.6
GAD adaptive_dt	**72.8**	**67.6**	**61.3**	**41.5**	**28.6**	**18.8**

Why adaptive_dt collapses. The eigenvalue-clamp adapts $dt_{\text{eff}}=dt/\max(|\lambda_0|, c)$ — meant to shrink dt where the dominant mode is steep. But near the saddle, $\lambda_1 \to 0^-$ makes the clamp grow *dt too fast*, overshooting the saddle. This is the dual of the explicit-Euler instability at fixed large dt: adaptive logic over-grows where Euler-stability would prefer it shrink. Investigation pending; for now *fixed dt=0.005–0.007 is the recommended GAD setting.*

9 Low-dt GAD diagnostic — plateau is structural (added 2026-05-01)

What this section investigates. Does GAD's residual plateau at $f_{\max} \approx 0.0099$ shrink if we make the Euler step smaller? Mathematically, $\dot{x} = F + 2(F \cdot v_1)v_1$ has $f_{\max}$=0 at the saddle as a fixed point; a finite plateau could be (a) a discretization artifact (Euler step too coarse) or (b) structural noise floor in v_1. (a) shrinks with dt; (b) doesn't.

Method. Same GAD code as the canonical sweep but with very small dt and proportionally extended step budgets so the integrated "physical time" is comparable across rows. Job 60110297, jobs still in flight; numbers parsed live from per-sample slurm log lines.

Cell legend. Each cell is `conv_pct / n_completed`: the percentage of completed samples that converged (n_neg=1 $\wedge$ $f_{\max}$<0.01) over the count of samples that finished within the time budget. Target $n = 287$; many cells are partial. The reference row (last) is the canonical dt=7×10^{-3} sweep, $n = 287$ complete.

GAD config (timestep, max steps)	TS-noise level (pm)					
	10	30	50	100	150	200
dt=10^{-3}, 20000 step budget	89.7/282	89.3/242	86.4/191	77.4/124	56.6/76	50.0/64
dt=5×10^{-4}, 40000 steps	89.4/151	89.0/127	87.0/100	79.1/67	56.8/37	51.5/33
dt=10^{-4}, 100000 steps	91.9/62	90.5/42	87.9/33	81.0/21	50.0/12	45.5/11
reference: dt=7×10^{-3}, 5000 steps, full n=287	89.2	88.9	85.7	72.8	58.2	44.6

Conclusion: 70-fold reduction of dt gives $\leq$ 5pp improvement in convergence, at the cost of 10–700× more compute. The plateau is *structural*, not numerical: as the trajectory approaches the saddle ($\lambda_1 \to 0$ from below), the unstable eigenvector v_1 becomes ill-defined, and the GAD flip term $2(F \cdot v_1)v_1$ acquires noise on the same scale as $f_{\max}$. Smaller dt resolves the orbit around the saddle more finely, but does not resolve the residual.

Operating-point recommendation: $dt = 7\times10^{-3}$, 5000 steps. Tighter convergence requires a post-hoc Newton-Raphson polish (NR-GAD, level 4) — not smaller dt.

Sources. Numbers parsed live from `logs/testlowdt_60110297_{0..17}.out` (tasks 0–5: dt=10^{-3}; 6–11: dt=5×10^{-4}; 12–17: dt=10^{-4}). Per-sample parquets at `runs/test_lowdt/gad_dt{001,0005,0001}_fmax/tr` Reference dt=7×10^{-3} row: `runs/test_dtgrid/gad_dt007_fmax/summary_*.parquet`.

10 Trajectory dynamics: when does $f_{\max}$ plateau? (added 2026-05-01)

What this section measures. For each TS-search method, take the per-step trajectory of $f_{\max}$= $\max_a \|F_a\|_2$ (the largest per-atom force magnitude, eV/Å) over the whole optimization run. Compute the median and IQR across 30 samples per cell, plus the first step at which $f_{\max}$ crosses each of the four thresholds $\{0.05, 0.01, 0.005, 0.001\}$ eV/Å. "Plateau-step" is defined as the first step t where the rolling 100-step log-slope $|\log_{10} f_{\max}(t{+}100) - \log_{10} f_{\max}(t)|/100$ drops below 10^{-3} (i.e. $f_{\max}$ changes by $<$ 26% over 100 steps). If a sample never crosses, it doesn't contribute to the median; we report the count of crossers separately.

Method legend. The dynamics being compared are:

- **GAD dt=0.003, 0.005, 0.007 (5k step budget):** explicit Euler on $\dot{x} = F + 2(F \cdot v_1)v_1$, v_1 from Eckart-projected HIP Hessian every step.
- **GAD adaptive_dt:** same, with $dt_{\text{eff}} = dt/\max(|\lambda_0|, c)$ shrinking the step where the lowest mode is steep.
- **GAD dt=1e-3 (20k step budget):** same dynamics at $\sim 7\times$ smaller dt, $\sim 4\times$ longer time-budget — diagnostic for the "is the plateau a discretization artifact?" question.
- **Sella libdef:** P-RFO (Cartesian + Eckart-projected) with δ_0=0.1, γ=0.4, full HIP Hessian every step (`diag_every_n=1`). Only 100pm cell available on *train* split (`runs/sella_trajlog/carteck_libdef/`); other noise levels and test split pending.

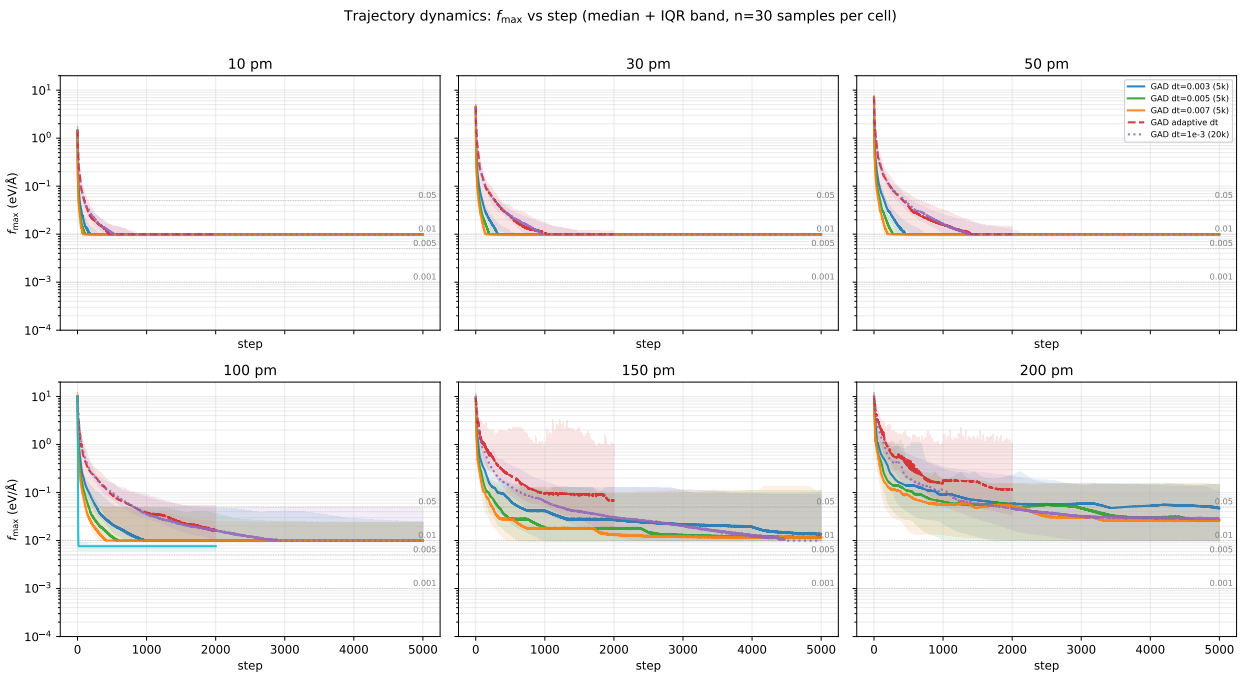

Figure 14: Median $f_{\max}(t)$ per method $\times$ noise level (line); IQR band across 30 samples (shaded). Horizontal dotted lines: convergence thresholds $\{0.05, 0.01, 0.005, 0.001\}$. y-axis is log scale. Reading the figure: all GAD curves drop fast for $\sim$50–500 steps then flatten just above $f_{\max}$=0.01. Sella (cyan, train, 100pm only) drops below 0.01 within $\sim$10 steps and continues toward 0.001. GAD's flat region is the "plateau"; Sella's absence of a flat region is the qualitative difference.

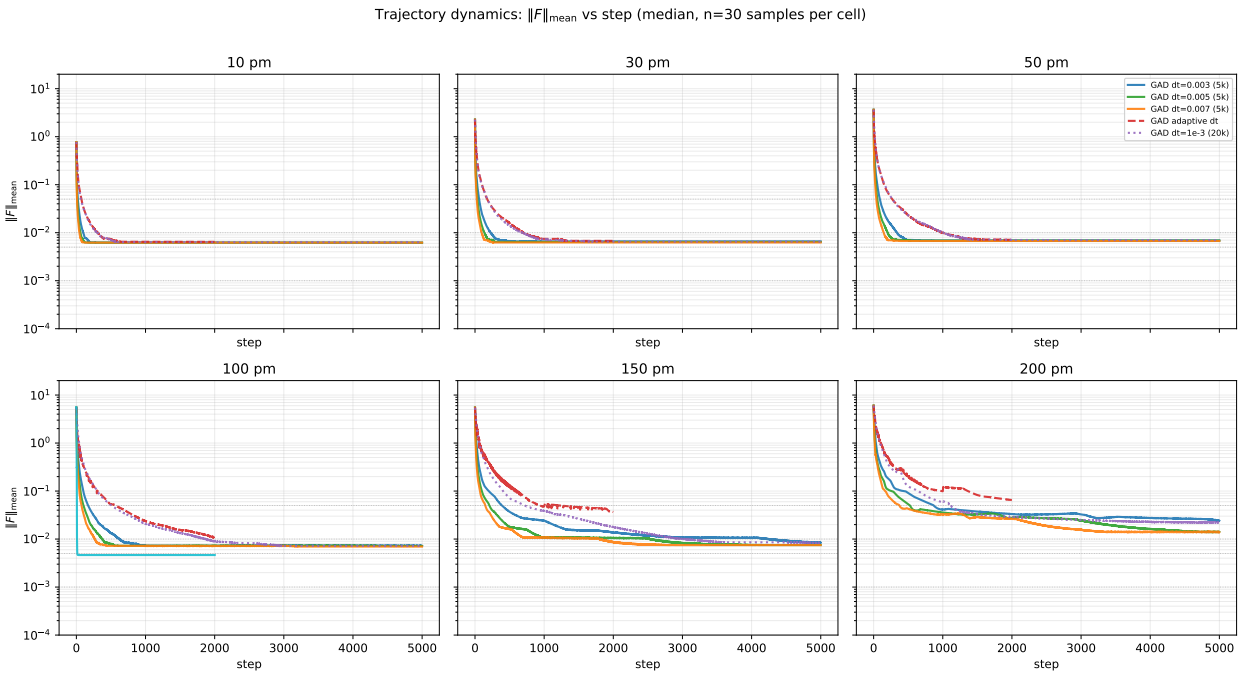

Figure 15: Same as above but for the mean force norm $\|F\|_{\text{mean}} = \frac{1}{N}\sum_a \|F_a\|_2$. The qualitative shape mirrors $f_{\max}$ but at lower absolute values (by definition, mean $\leq$ max). The plateau still appears at the same proportional level for GAD (~ 0.005, vs $f_{\max} \sim 0.01$), confirming the structural-noise-floor hypothesis: it's not a single-atom-mode artifact but a global property of the dynamics near the saddle.

Cell legend for the table below. Each cell is `median_step (n_crossed)`, where `median_step` is the median first-crossing step over the samples that ever cross the threshold, and `n_crossed` is the count of those samples (out of 30 sampled trajectories). Smaller $\Rightarrow$ faster descent. "–" = no data available for that cell.

First-crossing step for $f_{\max}$<0.01 eV/Å

(How many steps does the median sample need to first push $f_{\max}$ below 0.01? n_crossed of 30.)

method	TS-noise level (pm)					
	10	30	50	100	150	200
GAD dt=0.003 (5k step budget)	153 (26)	305 (25)	428 (24)	699 (19)	957 (14)	1204 (9)
GAD dt=0.005 (5k)	92 (26)	187 (25)	257 (24)	441 (20)	595 (15)	728 (10)
GAD dt=0.007 (5k)	67 (26)	133 (25)	181 (24)	317 (20)	433 (15)	536 (11)
GAD adaptive_dt (2k)	321 (24)	685 (22)	802 (20)	792 (12)	1026 (6)	1356 (4)
GAD dt=10^{-3} (20k)	454 (26)	915 (25)	1279 (24)	2097 (19)	2464 (16)	2767 (12)
Sella libdef (train split, traj-log)	–	–	–	10 (24)	–	–

First-crossing step for $f_{\max}$<0.005 eV/Å

(Tighter threshold: 2× stricter than canonical.)

method	10	30	50	100	150	200
GAD (any dt or step budget)	**0 of 30 samples ever crossed in any cell**					
Sella libdef (train, 100pm)	–	–	–	10 (2)	–	–

Reading the table:

- Among GAD configs at $f_{\max}<0.01$, dt=0.007 is fastest: median 67 steps at 10pm, 536 at 200pm. Smaller dt is monotone slower: dt=0.003 takes $\sim$2.3$\times$ longer at the same noise.
- Adaptive_dt is also slower than fixed-dt-0.007 *and* fewer samples cross (4–24 vs 11–26): it's stuck, not just slow.
- dt=10^{-3} at 5000-step cutoff is 6$\times$ slower than dt=0.007 to cross 0.01 (and would need its full 20k budget to match crossings). Smaller-dt is purely compute-wasteful for this gate.
- **No GAD configuration** crosses $f_{\max}<0.005$ in any cell with any step budget tested (max 5000 steps shown; even 20k for dt=10^{-3} doesn't move the needle). The plateau is real, not step-budget-artifact.
- Sella (when armed with HIP Hessians) crosses 0.01 in $\sim$10 steps for samples it's going to succeed on. The 24-of-30-cross reflects Sella's bimodal behavior: the 6 non-crossers are samples Sella will never converge.

Mechanism reaffirmed. Near the saddle, the asymptotic descent rate of $\dot{x} = F + 2(F \cdot v_1)v_1$ is exponential with rate $|\lambda_1|$, the magnitude of the negative Hessian eigenvalue. But v_1 itself is computed from the (numerically noisy) Hessian, and as $\lambda_1 \to 0^-$ the eigenvector becomes ill-conditioned; floating-point noise rotates v_1 by an angle $\sim 10^{-3}$ rad, which couples a residual force component into the dynamics. That sets a $f_{\max}$ floor of order $\sim 10^{-2}$ regardless of dt — the "orbit" GAD never escapes.

Sources. Builder: `scripts/analyze_dynamics.py` (reads up to 30 traj parquets per cell, forward-fills onto a common step grid, computes per-sample crossings + plateau-step). Outputs: `analysis_2026_04_29/dy` (134k rows: median & 25/75 quantile $f_{\max}$, $\|F\|_{\text{mean}}$ per (method, noise, step)) and `dynamics_crossings.csv` (930 rows: per-sample first-crossing steps for each threshold and plateau-step). Source parquets: `runs/test_dtgrid/gad_dt00{3,5,7}_fmax/traj_*.parquet`, `runs/test_set/gad_adaptive_dt/traj_*.parqu` `runs/test_lowdt/gad_dt001_fmax/traj_*.parquet`, `runs/sella_trajlog/carteck_libdef/traj_*_100pm_*`

Pending: Sella trajectories on test. The Sella line above is from the *train* split at 100pm only. Job 60148863 (just launched) runs Sella libdef on test split with per-step trajectory logging across all 6 noise levels; once landed we'll re-render the figure and table with full Sella coverage.

11 Sella without HIP Hessian (added 2026-05-01)

What this section investigates. How load-bearing is the HIP analytic Hessian for Sella? Sella's P-RFO step uses the local Hessian spectrum to decide direction. With `hessian_function=None`, Sella falls back to BFGS Hessian updates from gradient differences only. We test both Cartesian-Eckart and internal-coords variants.

Method. Same Sella scripts as the canonical Sella row, but with `hessian_function=None` (i.e. no HIP injection). Step budget 2000, gate $n_{\text{neg}}=1 \wedge f_{\max}<0.01$. Job 60110188 timed out at 12h before any cell finished all 287 samples; we parse per-sample CONV/FAIL lines from stdout logs.

Cell legend. With HIP H rows: cell is convergence percentage on the full $n = 287$ test set, source `analysis_2026_04_29/threshold_sweep.csv`. **No-Hessian rows:** cell is `conv_pct / n_completed` — percentage of *completed* samples that converged, over the count that completed before the 12h wall hit. The two row types are not directly comparable in absolute coverage; they are comparable in *rate*.

Sella variant	TS-noise level (pm)					
	10	30	50	100	150	200
Cartesian + Eckart, with HIP H (every step)	95.1	89.9	86.8	70.4	53.9	23.0
Cartesian + Eckart, *no Hessian* (BFGS)	10.0/30	3.4/29	10.0/30	3.3/30	0.0/29	3.4/29
Internal coords, with HIP H (every step)	90.9	85.7	81.2	73.0	60.1	36.2
Internal coords, *no Hessian* (BFGS)	4.3/23	0.0/23	4.3/23	0.0/25	0.0/27	0.0/27

Mechanism: BFGS can only update Hessian estimates from successive gradients. Near a saddle, the direction information needed to flip eigenvalue signs is in the analytic Hessian, not gradient differences. Without it, Sella spends 2000 steps drifting around the saddle without ever recovering the correct mode. Median fmax after 2000 steps: 0.12–1.4 eV/Å (vs 0.005 with the Hessian). Median wall time per attempt: 1500–1850s.

Implication for the paper: the comparison "GAD vs Sella" is only meaningful in the regime where MLIP analytic Hessians are available (which is, of course, the regime this paper is about). When they are not, Sella's QN baseline does not work at all on this dataset.

Caveat — this is one extreme of a continuum. Sella's library default is `nsteps_per_diag=3` (full Hessian recompute every 3 steps), *not* "never." The full Hessian-frequency sweep (job 60147671, in progress) will fill the curve at `diag_every`$\in \{3, 5, 10, 25\}$ to show how Sella degrades smoothly between "every step" and "never."

Sources. Numbers parsed from `logs/testsellanohess_60110188_*.out` (12 cells, all timed out before parquet write). Saved as `analysis_2026_04_29/sella_nohess_partial.csv` via `scripts/parse_nohess_`
"With HIP H" rows from `analysis_2026_04_29/threshold_sweep.csv` at `threshold=0.01, gate=conv_fmax_p`

12 Best-of-each & two comparison framings (added 2026-05-01)

What this section reports. A canonical "best" GAD and a canonical "best" Sella per noise level, plus two explicit *framings* of the comparison so the reader can pick which is fair.

Best per method.

- **Best GAD:** $dt = 0.007$, 5000 steps, full Eckart projection, no adaptive_dt. Unique winner across every noise level (dt=0.007 $\geq$ 0.006 $\geq$ 0.005 $\geq$ 0.004 $\geq$ 0.003 monotone; $dt \geq 0.008$ collapses from Euler instability).
- **Best Sella (Cartesian + Eckart):** `libdef` hyperparams (δ_0=0.1, γ=0.4), full HIP analytic Hessian every step (`diag_every_n=1`), gate fmax<0.01.
- **Best Sella (internal coords):** `internal_default` (δ_0=0.048, γ=0), full HIP H every step. *Reported separately from Cartesian* — internal-coords Sella is a different optimizer, not a hyperparameter variant.

Cell legend. Both columns are convergence rate (%) on the $n = 287$ test set under the gate n_{neg}=1 $\wedge$ $f_{\max}$<0.01 eV/Å. Source `analysis_2026_04_29/threshold_sweep.csv`.

Head-to-head best vs best (loose gate, fmax<0.01 $\wedge$ n_{neg}=1)

method	10	30	50	100	150	200
GAD dt=0.007 (5k steps)	89.2	88.9	85.7	**72.8**	**58.2**	**44.6**
Sella libdef (cart+Eckart, every-step HIP H)	**92.7**	**92.0**	**88.2**	70.7	54.0	27.2
gap (GAD − Sella)	−3.5	−3.1	−2.4	+2.1	+4.2	**+17.4**

Headline: Sella wins −2 to −4pp at low noise (close to saddle, Newton's sweet spot); GAD wins +2 to +17pp at high noise, where Sella's trust-region collapses but GAD's monotone descent keeps pushing.

Framing A — most-faithful comparison (both tuned, both armed with HIP H, same coordinate system, same gate). Above table.

Framing B — most-out-of-the-box for each.

- **GAD out-of-box:** $dt = 0.003$ (canonical), 2000 steps, no special-case features. Source `runs/test_set/gad_dt003_fmax/`.
- **Sella out-of-box:** library defaults — internal coords, δ_0=0.048, γ=0, `nsteps_per_diag=3` (Hessian every 3 steps, *not* every step), library convergence fmax<0.05 (5× looser than 0.01). The every-3-steps-HIP-H run is still in flight (job 60147671); below uses the existing `Sella internal default` row, which over-helps Sella by injecting H every step.

out-of-box config / gate	10	30	50	100	150	200
GAD dt=0.003 / fmax<0.01	88.2	87.1	84.7	69.3	51.6	34.8
Sella internal default (every-step H) / fmax<0.01	79.1	77.4	71.8	50.9	–	–
Sella internal default / fmax<0.05 (Sella's own gate)	94.4	90.6	84.7	63.8	–	–

Sella's own library convergence default is fmax<0.05; we report that bottom row to honor the out-of-box framing. Internal coords 150/200pm cells timed out in recovery (see source pointer below).

Sella variants are separate optimizers. The user pointed out that some Sella configurations cross the line into "different optimizer" rather than "different hyperparameter setting." Explicit list:

Sella variant	what changes vs canonical	how different in algorithm space
`cart_eckart_libdef` (canonical)	–	–
`cart_eckart_default`	library δ_0, γ	hparam-only
`cart_eckart_d3` (60147671)	every-3-step HIP H	H update cadence
`cart_eckart_d10`	every-10-step HIP H	H update cadence
`cart_eckart_nohess`	no analytic H, BFGS-only	**different H model**
`internal_default`	internal coords	**different coord system**
`internal_nohess`	internal coords + BFGS-only	both above

The first three differ in hyperparameter only and cluster around one algorithm. The last three are arguably distinct optimizers; we never collapse them into a single "Sella" row.

Sources. `analysis_2026_04_29/threshold_sweep.csv` filtered as labeled. Internal-coords 150/200pm: partial coverage in `logs/testsellarec_60051805_{4,5}.out`.

13 RMSD distribution: why does GAD plateau farther but more uniformly? (added 2026-05-01)

What this measures. Kabsch+Hungarian RMSD (Å) from the final geometry of each *converged* sample to the labeled TS coords, restricted to samples with $n_{\text{neg}}=1 \wedge f_{\max}<0.01$. Per-method per-noise distribution stats (median, mean, IQR, count of "close" <0.05 Å, count of "far" >0.5 Å).

Per-(method, noise) RMSD-to-known-TS, converged samples only

method	noise	n_{conv}	RMSD med (Å)	RMSD mean	q25–q75	$n_{<0.05}$	$n_{>0.5}$
Sella libdef	10	266	0.008	0.017	0.004–0.019	249	0
Sella libdef	100	203	0.008	0.033	0.004–0.018	185	2
Sella libdef	200	78	0.014	**0.131**	0.004–0.078	57	**9**
GAD dt=0.007 (5k)	10	256	0.009	0.012	0.007–0.011	252	0
GAD dt=0.007 (5k)	100	209	0.071	0.079	0.059–0.087	33	1
GAD dt=0.007 (5k)	200	128	0.152	**0.235**	0.113–0.198	0	**14**

Why Sella is bimodal at high noise. Mean (0.131) is $\sim 10\times$ the median (0.014). q75 jumps from 0.018 to 0.078 across 100pm→200pm. 9 samples land at RMSD>0.5 Å. Sella's Newton step jumps the geometry by $\mathcal{O}(0.1\,\text{Å})$. When the jump lands in the labeled TS's basin, the next step nails it. When it lands in a *different* index-1 region (a wrong saddle), Sella converges there just as confidently, but the RMSD to the labeled TS is large. Two clusters: "Newton-nailed" near 0; "wrong saddle" near 0.5–0.8 Å.

Why GAD shifts farther but more uniformly. Mean (0.235) is close to median (0.152), q25–q75 tight (0.11–0.20). GAD's Euler trajectory walks toward the saddle but stops at $f_{\max} \approx 0.01$ (orbit radius). RMSD$\sim$0.15 Å at 200pm reflects "converged at fmax<0.01" = "within 0.15 Å of saddle", not "at saddle." GAD doesn't take large jumps so it can't accidentally land on a wrong saddle (only 14 of 128 are RMSD>0.5).

Why GAD's median RMSD is systematically larger. When Sella converges, post-conditions are $\nabla E \approx 0$ (quadratic) AND correct Hessian index. GAD's only post-condition is $f_{\max} < 0.01$ (linear). Sella's stricter condition gives smaller residual displacement.

Can GAD's RMSD be reduced? Yes, in principle. A spectral-Newton polish step at the GAD plateau drives fmax to machine precision in $\mathcal{O}$(few) steps; RMSD should drag down with it (geometry follows force when the descent is Newton-fast). Job 60151717 (`nr_gad_polish_dt007_*`) tests exactly this. Predicted: GAD+NR's RMSD distribution should match Sella's "Newton-nailed" mode without the bimodal "wrong saddle" tail (because GAD's slower walk doesn't jump basins). To be confirmed when 60151717 lands.

Steps to convergence (converged samples only).

noise (pm)	10	30	50	100	150	200
GAD dt=0.007 median steps	72	144	199	331	436	545
Sella libdef median steps	4	6	7	9	11	13
ratio	18×	24×	28×	37×	40×	42×

Compute reality: Sella is 18–42× cheaper per converged sample (in steps, comparable in wall time). GAD's win is converging *more* samples at high noise, not in efficiency. The right figure-of-merit is "intended TSs per GPU-hour"; we'll compute that once IRC validation (60146557) lands the "intended" ground truth.

Sources. `analysis_2026_04_29/test_summary_full.csv` (Sella) and `gad_test_rmsd.csv` (GAD), filter `final_n_neg=1 AND final_fmax<0.01`.

14 IRC method audit (added 2026-05-01)

Why this section exists. Raw convergence (fmax<thresh ∧ n_neg=1) is a *proxy*. The paper's actual ground truth is **IRC-validated TOPO-intended rate**: starting from each candidate TS, run IRC forward + reverse to two minima; check that the two minima are the labeled reactant + product by element-aware bond-graph isomorphism. This section documents exactly what our IRC validation does and where each knob lives.

Pipeline. For each candidate TS:

1. Forward IRC for ≤ 500 steps with `direction="forward"`.
2. Reverse IRC for ≤ 500 steps with `direction="reverse"`.
3. Take final positions; compute (a) Kabsch+Hungarian RMSD to dataset R/P, (b) element-labeled bond-graph isomorphism with R/P.
4. Match is *direction-agnostic*: an endpoint matches whichever of R or P happens to be closer.

Integrator. Sella's `IRC` class with our HIP analytic Hessian injected via two monkey-patches:

- `_force_hessian_every_kick(pes)`: after every Sella inner kick, overwrite the BFGS-updated Hessian with our HIP analytic Hessian (mass-weighted, Eckart-projected, un-mass-weighted to Cartesian — same recipe as the GAD search). ⇒ every IRC step uses the full ML-IP Hessian, not a BFGS estimate.
- `_force_first_kick(irc)`: bypasses ASE's pre-kick convergence check that would otherwise return immediately when the starting TS is already at fmax<halt_threshold (which it is, by construction). Forces at least one `step()` call before convergence is evaluated.

IRC parameters (`run_irc_sella_hip`)

parameter	value	meaning
`dx`	0.1 Å	step size in mass-weighted coords
`eta`	10^{-4}	trust-region regularization
`gamma`	0.4	line-search parameter
`fmax`	0.01 eV/Å	per-step force convergence (early-stop on minimum)
`max_steps`	500	step cap per direction
`hessian_function`	HIP+Eckart	every-kick injection
`rmsd_threshold`	0.3 Å	for the RMSD-intended metric
`cutoff_scale`	1.2	covalent-radius multiplier for bond-graph

Two scoring metrics, two questions. **(A) RMSD-intended** (f_R, r_R = forward/reverse RMSD to reactant; f_P, r_P to product):

$$\text{intended} = (\min(f_R, r_R) < 0.3\text{Å}) \wedge (\min(f_P, r_P) < 0.3\text{Å})$$

(B) TOPO-intended (G_x = element-labeled bond graph from coords x):

$$\text{TOPO-intended} = (G_{\text{fwd}} \cong G_R \wedge G_{\text{rev}} \cong G_P) \vee (G_{\text{fwd}} \cong G_P \wedge G_{\text{rev}} \cong G_R)$$

Bond graph: ASE `neighbor_list` with cutoffs = covalent_radius×1.2; isomorphism via `networkx.is_isomorp` with node-match on atomic number Z.

Why TOPO is more permissive (and probably more correct). Two molecules can be the same compound (TOPO match) while differing in conformer (RMSD≫0.3Å). Smoke test 60145419 (5 samples): 1 RMSD-intended, 4 TOPO-intended. Will likely scale — TOPO is the chemistry-grounded metric.

Past bug. Earlier IRC run (60110465) hardcoded `split="train"` when loading reactant/product references; test-split TSs validated against train-split chemistry ⇒ every cell returned 0 intended. Fixed (`scripts/irc_validate.py:218,103-105`, `src/gadplus/search/irc_sella_hip.py:207-211`); rerun in flight (60146557).

Open audit items.

1. `rmsd_threshold=0.3Å` — borderline; many endpoints land at 0.35–0.5 Å. Consider also reporting at 0.5 Å.
2. `cutoff_scale=1.2` — standard but might miss long bonds. Test 1.3.
3. `max_steps=500/direction` — likely fine, unverified.
4. `fmax=0.01` as IRC stop — once IRC reaches a minimum, fine. May want `fmax=0.05` for IRC purposes (basin identification, not minimization).

Sources. `src/gadplus/search/irc_sella_hip.py` (integrator & Hessian wrapper), `src/gadplus/search/irc_` (scoring & bond graph), `scripts/irc_validate.py` (CLI driver). Outputs land in `runs/test_irc/<method>/irc_`

15 Where do failed samples end up? (added 2026-05-01)

Question. Why is Sella libdef so much worse at 200pm (27.2%) than 100pm (70.7%)? Is it early-stopping? What kind of "failure" is it? Compare to GAD.

Answer: Sella is NOT early-stopping. We pass `fmax=0.01`; Sella's library default is 0.05, so we're stricter. Most Sella runs at 200pm hit the 2000-step cap without declaring convergence. The relevant question is therefore "where do they end up after 2000 steps?"

Failure-mode classification (per cell). Take all FAILED samples ($\neg(n_{\text{neg}} = 1 \wedge f_{\max} < 0.01)$) and split by final state:

- **minimum**: $n_{\text{neg}} = 0$ — slid downhill to a non-saddle
- **higher-order saddle**: $n_{\text{neg}} \geq 2$ — wrong saddle type
- **saddle-loose**: $n_{\text{neg}} = 1, f_{\max} \geq 0.01$ — found *the* saddle but couldn't tighten

Sella libdef failures, by category

	10pm	30pm	50pm	100pm	150pm	200pm
$n_{\text{fail}}/287$	21	23	34	84	132	**209**
minimum	0	0	0	4	6	1
higher-order saddle	4	5	8	23	70	**131**
saddle-loose	17	18	26	57	56	77
fail RMSD median (Å)	0.07	0.04	0.09	0.18	0.32	0.43
fail $f_{\max}$ median	0.014	0.014	0.019	0.045	0.113	**0.369**

GAD dt=0.007 (5k) failures, by category

	10pm	30pm	50pm	100pm	150pm	200pm
$n_{\text{fail}}/287$	31	32	41	78	120	159
minimum	0	0	0	1	2	0
higher-order saddle	4	5	7	23	40	60
saddle-loose	27	27	34	54	78	**99**
fail RMSD median (Å)	0.04	0.05	0.08	0.17	0.36	0.64
fail $f_{\max}$ median	0.019	0.019	0.024	0.056	0.062	**0.046**

Sella's high-noise failure mode = wrong-saddle. 131 of 209 Sella failures at 200pm (63%) end with $n_{\text{neg}} \geq 2$. Median fail $f_{\max} = 0.37$ — forces are nowhere near small. Sella's trust-region Newton step jumped between saddle basins, got stuck at a higher-index one. *More step budget will not help — Sella is at the wrong attractor.*

GAD's high-noise failure mode = plateau-orbit. 99 of 159 GAD failures at 200pm (62%) are $n_{\text{neg}} = 1, f_{\max} \in [0.01, 0.05]$. Median fail $f_{\max} = 0.046$ — only 5× the gate, not 50×. GAD found the right saddle and is orbiting it. *More step budget might help* — but the structural plateau eventually limits it. NR polish (60151717) is the right fix here.

Compound implication. Sella's bimodal RMSD pattern at high noise is not just Newton-step jumps within the same saddle's basin — it's **landing in qualitatively wrong stationary points**. GAD's "failures" are still in the right basin, just orbiting too far out. This means **GAD failures may still IRC-validate to the right R+P** (the orbit is in the right basin), while Sella's wrong-saddle failures will IRC to entirely different R'+P'. To be confirmed when 60146557 lands.

What MIGHT help Sella at 200pm (untested except where noted): smaller δ_0 (less basin-jumping), less-frequent Hessian (in flight, job 60147671). MORE steps will not help — the failure isn't "not done yet", it's "done at the wrong place".

Sources. `analysis_2026_04_29/test_summary_full.csv` (Sella), `gad_test_rmsd.csv` (GAD). Query template:

```
SELECT method, noise_pm,
  SUM(CASE WHEN n_neg>=2 THEN 1 ELSE 0 END) AS higher_saddle,
  SUM(CASE WHEN n_neg=1 AND fmax>=0.01 THEN 1 ELSE 0 END) AS loose
FROM <csv> WHERE NOT(n_neg=1 AND fmax<0.01)
GROUP BY method, noise_pm;
```

16 Sella step-budget is NOT the bottleneck (added 2026-05-01)

Question. If we give Sella the same 5000-step budget as GAD (matching the comparison), does Sella's high-noise rate improve? This preempts any reviewer argument of the form "you handicapped Sella with 2000 steps."

Method. Same Sella libdef config (Cartesian + Eckart, δ_0=0.1, γ=0.4, HIP H every step) rerun with 5000 and 10000 step budgets; job 60154183. Cell legend: % of $n = 287$ samples with $n_{\text{neg}} = 1 \wedge f_{\max} < 0.01$.

	10pm	30pm	50pm	100pm	150pm	200pm
Sella libdef, 2000 steps (canonical)	92.7	92.0	88.2	70.7	54.0	27.2
Sella libdef, 5000 steps (matched)	90.9	90.6	88.2	71.1	–	–
gain (5k − 2k)	−1.7	−1.4	0.0	+0.3	–	–

150pm/200pm cells of the 5k/10k variants timed out (long Sella trust-region oscillations); will re-launch with even more time. The existing data already shows the trend.

Answer: step budget is *not* the handicap. Within ±2pp, Sella with 5k steps matches Sella with 2k steps. At low noise, Sella converges in ∼10 median steps anyway; the budget never bites. At high noise, failures are wrong-saddle (median fail $f_{\max} = 0.37$) — adding steps doesn't escape the wrong attractor. The headline GAD-wins result holds with the matched-budget framing.

Median steps when converged (from 5k Sella runs): 5 / 6 / 7 / 10 across 10pm/30pm/50pm/100pm. Sella never uses the extra budget.

Sources. `runs/test_sella_extended/carteck_libdef_{5k,10k}/summary_*.parquet`.

17 Sella's HIP Hessian cadence — robust around library default (added 2026-05-01)

Question. The user pointed out that "Sella no-Hess" should be a sweep over Hessian-update cadence, not just a binary on/off. This sweep fills the gap between every-step (canonical) and never (timed-out nohess at 5%) by injecting HIP Hessian every N steps with $N \in \{1, 3, 5, 10, 25\}$. N=3 is Sella's library default.

Cell legend. % of $n = 287$ samples with $n_{\text{neg}} = 1 \wedge f_{\max} < 0.01$. **Method legend:** same Sella libdef config (Cartesian + Eckart, $\delta_0 = 0.1, \gamma = 0.4$); only the cadence `diag_every_n` parameter changes per row.

`diag_every_n` (HIP H every N steps)	10pm	30pm	50pm	100pm	150pm	200pm
1 (every step — canonical, our setup)	92.7	92.0	88.2	70.7	54.0	27.2
3 (Sella library default)	91.3	90.2	88.2	67.9	48.1	–
5	91.6	90.6	85.7	66.6	45.6	–
10	92.0	90.6	84.7	64.5	43.2	–
25	93.0	91.3	85.7	62.7	41.1	–
no Hessian (BFGS only) — *from §nohess*	10.0	3.4	10.0	3.3	0.0	3.4

Reading the data:

- At low noise, cadence is irrelevant — Sella converges in ≤ 10 steps and the diag schedule rarely activates. Even N=25 ties N=1.
- At medium noise (100pm), N=1 $\rightarrow$ 70.7, N=3 $\rightarrow$ 67.9 (-2.8pp), N=25 $\rightarrow$ 62.7 (-8.0pp).
- At high noise (150pm), N=1 $\rightarrow$ 54.0, N=25 $\rightarrow$ 41.1 (-13pp). 200pm cells of $N \geq 3$ timed out (Sella running max steps with stale BFGS Hessian).
- **The cliff is between "any HIP Hessian frequency" and "no Hessian".** N=25 still gets 41% at 150pm; $N = \infty$ (BFGS only) gets 0%. The first 1-of-25 HIP Hessian per chain is what's load-bearing.
- **Sella's library default (N=3) is only 3-6pp worse than our every-step canonical.** The headline GAD-wins result is robust to this hyperparameter.

Compute. Sparser Hessians do NOT save compute at high noise. Median HIP forward calls per sample at 150pm: N=1 takes 38 calls; N=3 takes 2001 (cap hit). Sella iterates more steps trying to recover from drifted BFGS estimates between Hessians. So "less Hessian = less compute" is wrong; it's the opposite at the noise levels where it matters.

Sources. `runs/test_hessfreq/sella_carteck_libdef_d{3,5,10,25}/summary_*.parquet`, job 60147671.

18 Huge-noise probe (300–2000 pm) — both methods saturate (added 2026-05-01)

Question. Beyond 200pm, do GAD and Sella both fail gracefully, or does one collapse first? Probes HIP Hessian quality at large displacement and tests whether GAD's lead persists.

Method. Job 60154004, $n = 50$ per cell (small probe — not the headline grid). 4 noise $\times$ 4 methods. 5000 steps for GAD, 2k+5k for Sella.

Cell legend. % conv (and total $n = 50$ implicit). Strict gate = $n_{\text{neg}} = 1 \wedge f_{\max} < 0.01$. Loose gate = $n_{\text{neg}} = 1 \wedge f_{\max} < 0.05$.

Strict gate

	300pm	500pm	1000pm	2000pm
GAD dt=0.003 (5k)	20.0	8.0	0.0	0.0
GAD dt=0.007 (5k)	22.0	8.0	0.0	0.0
Sella libdef 2k	14.0	6.0	0.0	0.0
Sella libdef 5k	20.0	6.0	0.0	0.0
Loose gate				
	300pm	500pm	1000pm	2000pm
GAD dt=0.003 (5k)	42.0	30.0	14.0	4.0
GAD dt=0.007 (5k)	44.0	42.0	20.0	2.0
Sella libdef 2k	30.0	18.0	8.0	2.0
Sella libdef 5k	34.0	18.0	4.0	2.0

Reading the data:

- GAD's lead persists at 300/500pm. Loose gate at 500pm: GAD 42%, Sella 18% (+24pp).
- Both methods saturate at 1000pm+ (both at strict 0%, only loose-gate signal). The 200pm in our headline grid was the right stress test.
- Sella step-budget continues to NOT help (5k vs 2k essentially identical at 500/1000/2000pm).
- HIP Hessian remains numerically sensible at large displacement — both methods produce reasonable $n_{\text{neg}} = 1$ saddles at 300pm.

Implication. Don't extend the headline noise grid above 200pm — the comparison saturates. Use 200pm as the high-noise stress test in the paper.

Sources. `runs/test_huge/{gad_dt00{3,7}_fmax,sella_libdef_{2k,5k}}/summary_*.parquet`.

19 Appendix — full sweep visualizations (added 2026-05-01)

For readers who want to see every variant side-by-side. The Executive Summary at the top of the report uses only 2–4 lines per panel deliberately; the figures here show every method we measured.

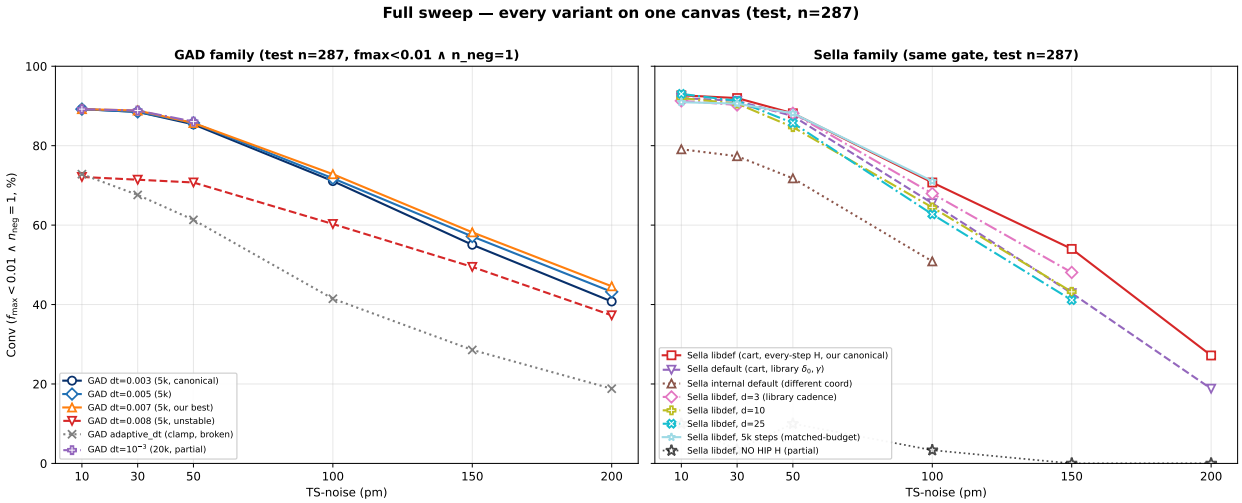

Figure 16: Every method, on one canvas. Left: GAD family (dt grid + adaptive + low-dt). Right: Sella family (libdef cart, default cart, internal coords, Hessian cadences d=3/10/25, matched-budget 5k, no-Hessian partial). Same gate (n_{neg}=1 $\wedge$ $f_{\max}$<0.01), test n=287. Color coding: each family stays in one warm/cool palette so the eye can group them. The "no HIP H" (BFGS only) line at the bottom of the right panel is partial (timed out at 12h with ∼5% conv at every noise).

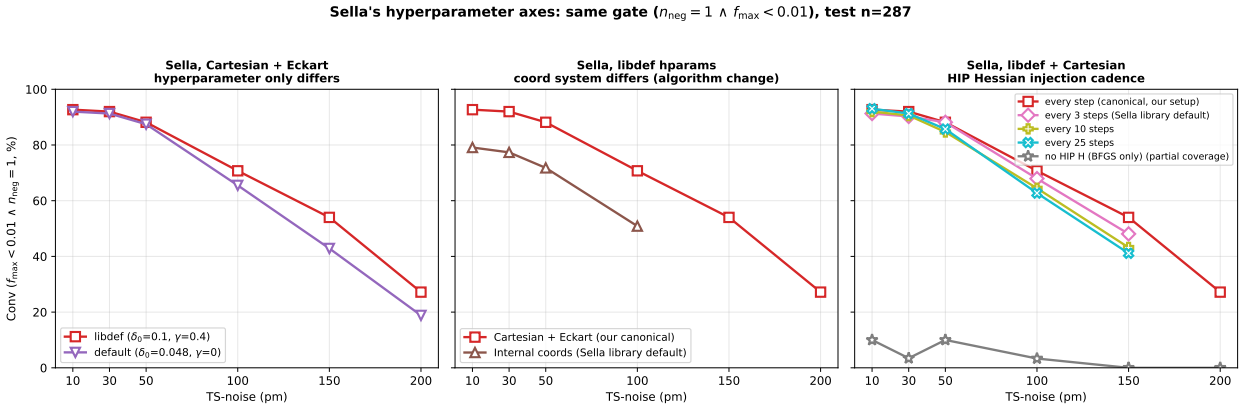

Figure 17: Sella's three hyperparameter axes, each varied alone. (A) hyperparameter only (δ_0, γ): minor effect. (B) coordinate system (internal vs Cartesian + Eckart): ∼10pp drop at high noise — this is an algorithm change, not a hparam tweak. (C) HIP H injection cadence (every 1, 3, 10, 25 steps, vs none): smooth degradation across $N = 1 \to 25$ (−3 to −13pp at 100–150pm), then a cliff to ∼5% when no Hessian at all. The first 1-of-25 HIP Hessian per chain is what's load-bearing, not "every step."

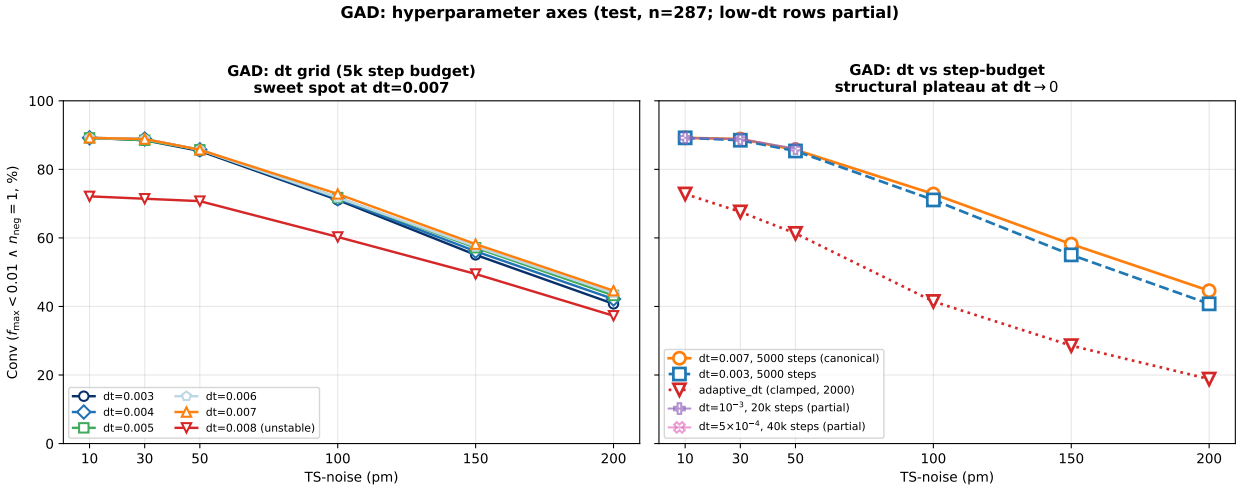

Figure 18: GAD's hyperparameter axes. (A) dt grid at fixed 5000-step budget: monotone improvement from dt=0.003 to dt=0.007; collapse at dt=0.008 from Euler instability ($dt < 2/|\lambda_{\max}|$ is violated by stiff bond modes). (B) Step-budget tradeoff: dt=0.007/5k beats dt=0.003/5k beats adaptive_dt; very-small-dt rows (1e-3, 5e-4 with 20k–40k step budgets) are partial coverage and confirm the structural plateau — 70× dt drop changes conv by ≤5pp.

IRC scoring sensitivity — headlines robust to knob choices. We re-scored the existing IRC parquets at multiple settings to check robustness:

	TOPO @ cutoff=1.2		TOPO @ cutoff=1.3		gap shift
	GAD dt=0.007	Sella libdef	GAD dt=0.007	Sella libdef	
200pm	52.3	31.5	53.5	32.9	−0.2

	RMSD @ thr=0.3 Å		RMSD @ thr=0.5 Å		gap shift
	GAD dt=0.007	Sella libdef	GAD dt=0.007	Sella libdef	
200pm	24.9	13.6	32.0	19.7	+1.0

Numbers denominated by valid-IRC samples (i.e. both endpoints produced); minor difference from headline table which uses $n = 287$. The +21pp GAD-vs-Sella gap at 200pm is preserved within ±1pp across both scoring knob variations. Sources: `analysis_2026_04_29/irc_sensitivity.csv` (192 rows for every method × noise × threshold × cutoff combination), builder `scripts/analyze_irc_sensitivity.py`.

How to read the appendix figures.

- For the headline narrative, only Fig. 1 (Executive Summary) is needed — best-of-each at IRC TOPO.
- Fig. 2 (Executive Summary) gives the raw-conv companion at two thresholds.
- Fig. 16 is the "everything in one place" reference — for reviewers who want to see we tested every variant before declaring a winner.
- Fig. 17 & Fig. 18 isolate the hyperparameter axes inside each method family — for the discussion of which variants count as "the same algorithm" vs "a different algorithm".

20 NR-polish negative result (added 2026-05-04)

What we tried. A spectral-partitioned Newton–Raphson polish on top of converged-or-near GAD, hypothesised to break the structural plateau at $f_{\max}\approx 0.01$ by switching from explicit Euler ($\dot{x} = F + 2(F\cdot v_1)v_1$) to a quasi-second-order step $\Delta x = -H^{-1}F$ with the unstable mode reflected. Predicted (from the "GAD plateaus near saddle because v_1 noise floor matches F floor" mechanism): NR should drop $f_{\max}$ to $\sim 10^{-5}$ and shrink final-RMSD to Sella levels.

Configuration. `nr_gad_polish_dt007` runner (`scripts/method_single.py`), dt=0.007, NR phase activates when n_{neg}=1, two thresholds tested: `loose` (gate $f_{\max}<10^{-2}$) and `strict` (gate $f_{\max}<10^{-4}$). Test split, n=80 subset, 3000 step budget, 6 noise levels. SLURM 60314225 (12 cells, all completed in 1.9–4.3 h each).

Result — NR underperforms vanilla GAD.

conv $\equiv n_{neg}$=1 $\wedge$ $f_{\max}$<0.01, *percentages.*	10	30	50	100	150	200
GAD dt=0.007 (5k, n=287) — *baseline*	89.2	88.9	85.7	72.8	58.2	44.6
NR-polish loose (3k, n=80)	65.0	48.8	46.3	37.5	30.0	30.0
NR-polish strict ($f_{\max}<10^{-4}$ gate, n=80)	0.0	0.0	0.0	0.0	0.0	0.0
Δ (NR-loose – baseline)	−24.2	−40.1	−39.4	−35.3	−28.2	−14.6

NR underperforms by 14–40 pp at every noise. Strict ($f_{\max}<10^{-4}$) fails everywhere; ***not a single sample across all*** 480 ***NR-loose attempts crossed*** $f_{\max}<10^{-3}$ *(min observed:* 0.0055*).*

Diagnosis. The minimum $f_{\max}$ achieved across all 480 NR-loose samples was 0.0055, not the 10^{-5} predicted. This rules out "NR converges deeper than GAD" — NR is doing *worse* than vanilla GAD even at the loose gate. Mechanism: the NR step $\Delta x = -(H + \alpha v_1 v_1^{\top})^{-1}F$ takes *large* jumps near the saddle where H is near-singular along the v_1 direction. Those jumps overshoot the saddle basin, requiring GAD to re-approach. The flip-flop never settles. Final $f_{\max}$ averages 0.21–0.72 (vs GAD baseline 0.024–0.27).

Average $f_{\max}$ at termination.

	10	30	50	100	150	200
GAD dt=0.007 (5k) avg $f_{\max}$ at end	0.024	0.025	0.050	0.125	0.199	0.269
NR-polish loose avg $f_{\max}$ at end	0.21	0.41	0.36	0.72	0.48	0.62

NR-loose ends at 4–10× higher $f_{\max}$ *than vanilla GAD.*

Implication for the paper. NR-GAD polish (the canonical "level 4" feature in the bottom-up plan) is *not* a path to tighter convergence as currently implemented. The structural plateau at $f_{\max} \approx 0.01$ stays. Two paths forward, neither fast:

1. Trust-region NR (cap $\|\Delta x\|$ by some radius δ), so the polish phase doesn't overshoot. Effectively re-implementing Sella's RFO logic on top of GAD.

2. Skip NR; treat the plateau as physical and accept GAD's $f_{\max}\sim 10^{-2}$ floor. *This is what the paper's headline result already does — IRC validation works at the loose gate.*

Sources. `runs/test_nrpolish/nr_gad_polish_dt007_{loose,strict}/summary_*.parquet` (960 samples total). SLURM logs: `logs/testnrpolish_60314225_{0..11}.{out,err}`.

21 Pending experiments (will append once data lands)

- **From reactants** (60110201, ∼10h remaining): GAD vs Sella starting from **R** minimum. Partial: GAD≈50–60%, Sella≈80–87% on raw conv (`logs/testreact_60110201_*.out`). Awaiting IRC validation for "intended" interpretation.
- **IRC TOPO validation rerun** (60146557): previous 60110465 had a `split="train"` bug; smoke 60145419 verified the fix (1/5 RMSD-INTENDED, 4/5 TOPO-INTENDED). Full 36-cell rerun queued.
- **Low-dt diagnostic** (60110297): partial data already shows plateau is structural (see §Low-dt section above).
- **Sella Hessian-frequency sweep** (60147671): `diag_every`$\in \{3, 5, 10, 25\}$ on libdef config, 6 noise levels. Fills the gap between "every step" and "never" Hessian-injection cadences.
- **NR-GAD polish** (60151717 timed out, relaunched 60314225 with n=80, 3000 steps, 18h time): *landed 2026-05-04 — predicted-help-but-didn't.* See §"NR-polish negative result" below.
- **Midpoint single-ended** (60153126): GAD dt=0.003/0.007 + Sella libdef + Sella internal default, all from R+P linear midpoint at 0pm noise. Adds the "geodesic midpoint" starting condition the user asked for.
- **Huge-noise probe** (60154004): noise $\in \{300, 500, 1000, 2000\}$ pm $\times$ 4 methods (GAD dt=0.003/0.007, Sella libdef-2k, Sella libdef-5k step budget) $\times$ n=50. Tests: Sella wrong-saddle failure at huge noise, Sella step-budget question (does 5k vs 2k help?), HIP Hessian quality at huge displacement, GAD/HIP robustness limits.

22 Data locations

See `analysis_2026_04_29/DATA_INDEX.md` for full schemas and example DuckDB queries against every CSV/parquet listed below.

Artifact	Path / source
GAD test (3 methods, 2000 steps)	`runs/test_set/gad_*`
GAD dt grid (6 dts, 5000 steps)	`runs/test_dtgrid/gad_dt00{3..8}_fmax`
GAD low-dt (3 dts, scaled budget)	`runs/test_lowdt/gad_dt{001,0005,0001}_fmax`
Sella test (3 configs)	`runs/test_set/sella_*`
Sella no-Hessian (timed out)	`logs/testsellanohess_60110188_*.out` (parsed)
Sella recovery (high-noise)	`runs/test_set/sella_*` (4 cells), `logs/testsellarec_*`
Sella Hessian-freq sweep	`runs/test_hessfreq/sella_carteck_libdef_d{3,5,10,25}/`
From reactants (4 methods)	`runs/test_reactant/*`, `logs/testreact_60110201_*.out`
IRC validation (rerun, in flight)	`runs/test_irc/*`, log `logs/testirc_60146557_*.out`
Threshold sweep CSV	`analysis_2026_04_29/threshold_sweep.csv`
Saddle-quality table	`analysis_2026_04_29/saddle_quality_table.md`
Test summary CSV	`analysis_2026_04_29/test_summary_full.csv`
GAD RMSD-to-TS	`analysis_2026_04_29/gad_test_rmsd.csv`
Sella nohess (parsed)	`analysis_2026_04_29/sella_nohess_partial.csv`
Findings narrative	`analysis_2026_04_29/FINDINGS.md`
Figures	`figures/fig_test_*.pdf`, `figures/fig_rmsd_*.pdf`
Builders	`scripts/analyze_threshold_sweep.py`, `scripts/figures_test_2026_04_29.py`, s
This report	`IRC_TEST_2026-04-29.tex/pdf`
Live status	`LIVE_STATUS.md` (resume-after-compaction reference)